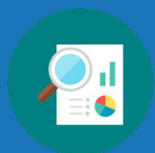
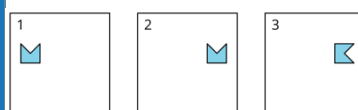


Unit 4 Overview: Rigid Motions and Congruence

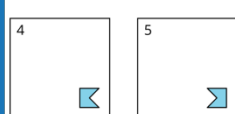
Unit Narrative



In this unit, students extend their prior learning of transformations to now include algebraic descriptions, justifying congruence, and more extensive use of the coordinate plane. Students are more formally introduced to the definition of congruence in terms of rigid motions, which was explored in Unit 1. Here is a sample of what activating prior knowledge can look like from Unit 1 to Unit 4:



Unit 1 Sample (Lesson 3)

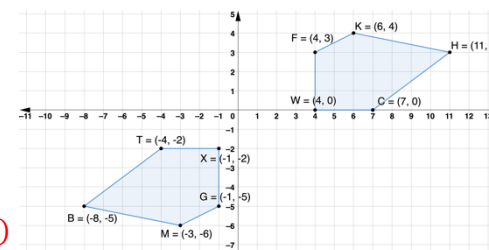


Movement from...	Description
Frame 1 to Frame 2	Slide to the right
Frame 2 to Frame 3	Turn to the right
Frame 3 to Frame 4	Slide down
Frame 4 to Frame 5	Turn the other way



Unit 4 Sample (Lesson 7)

$$\begin{aligned}(x, y) &\rightarrow (-x, y) \\(x, y) &\rightarrow (x, -y) \\(x, y) &\rightarrow (x + 3, y - 2)\end{aligned}$$



Students begin this unit by exploring rigid transformations (translations, rotations, and reflections) and writing their algebraic descriptions. Notice the progression from written descriptions in Unit 1 to algebraic descriptions in Unit 4 (sample above). Scaffolding for the transition from written to algebraic descriptions includes the use of multiple representations and methods for identifying and producing sequences of transformations. The unit begins with understanding the algebraic description for one transformation before extending to multiple transformations provided in a sequence. They will continue with sequences of transformations that are provided in both written and algebraic descriptions to explore isometry and justify congruence.

Section 2 takes what students have learned in Section 1 and applies it to a formal introduction to the definition of congruence in terms of rigid motions first introduced in Unit 1. Students will extend the concept of isometry from Unit 1 to relate it to figures on the coordinate plane, which then leads to the formal definition of congruence in terms of rigid motions. They will use the definition to justify the congruence of two figures by showing the mapping of the rigid motions. The unit closes with a Card Sort activity where students will match graph cards with their corresponding written description and algebraic description cards.

Unit At-A-Glance

	Benchmarks Addressed	Lesson	
Section 1: Sequence of Transformations	MA.912.GR.2.1 MA.912.GR.2.2 MA.912.GR.2.3 MA.912.GR.2.5	Lesson 1	Algebraic Descriptions for Translations and Rotations
		Lesson 2	Algebraic Descriptions for Reflections
		Lesson 3	Sequences of Transformations
		Lesson 4	Written Descriptions of a Sequence of Transformations
		Lesson 5	Algebraic Descriptions of a Sequence of Transformations
Section 2: Geometric Congruence	MA.912.GR.2.3 MA.912.GR.2.6	Lesson 6	Isometry
		Lesson 7	Justifying Congruence
		Lesson 8	Putting It All Together



Differentiation

Universal Design for Learning™ © CAST, 2008

Engagement <i>The “WHY” of learning</i>	Visible Learning, Processing, and Anchor Charts: Students will be engaging in activities throughout the unit that stimulate all three phases of learning—surface, deep, and transfer. These phases directly contribute toward the students’ development of “personal coping skills and strategies” while promoting self-regulation as students engage with the content. Read more about the phases of Visible Learning in the section below.
Representation <i>The “WHAT” of learning</i>	- This unit naturally allows for active participation, exploration, and experimentation through the manipulation of images. Provide a variety of resources that students can sample and choose from. - Patty paper - Physical geometric shapes - Geoboards and rubber bands - Reflective devices - Technology (Geogebra and Desmos)
Action & Expression <i>The “HOW” of learning</i>	- Consider an interdisciplinary lesson with an art teacher in which students explore the use of transformations in paintings and other forms of art. Use one class period to explore the artistic aspects to engage students in the concept followed by another class period quantifying its properties. - Lesson 2: Squish Painting (Wrap-Up) – Students can map, scale, and provide transformations for the squish. - Lesson 4: Kaleidoscope (Warm-Up) – Students can create their own kaleidoscopes using various transformations.

ELL Information

Teachers can have students create a V-I-S chart (Vocabulary Improvement Strategy). When students are asked to provide an illustration (picture), some examples can be: rotation - Earth, reflection - mirror, translation - elevator, and dilation - eyes.

V.I.S. CHART (Vocabulary Improvement Strategy)

Word	Meaning (use context clues)	Sentence (use one from the text, if available)	Personal Clue (drawing or a synonym)

Encourage students to make dual-language flashcards of the vocabulary words, properties, and theorems, including diagrams where appropriate.

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support” below:

- Congruent
- Reflection
- Rigid transformation
- Rotation
- Translation

Additional Differentiated Support

Scaffolding Resources

Visible Learning and Mathematics

John Hattie’s work in conceptual understanding and optimizing student learning offers phases or stages for students transferring their learning from the surface to long-term retention and assimilation. Unit 4 offers a variety of opportunity for students to actively engage with each phase both intra-lesson and between lessons.



Notice this sample from Lesson 2 in which students transition through each phase, with this approach reflected in every lesson throughout the unit.

Surface learning phase: When—through carefully constructed experiences—students explore new concepts and make connections to procedural skills and vocabulary that give shape to developing conceptual understandings.

Deep learning phase: When—through the solving of rich, high-cognitive tasks and rigorous discussion—students make connections among conceptual ideas, form mathematical generalizations, and apply and practice procedural skills with fluency.

Transfer phase: When students can independently think through more complex mathematics and can plan, investigate, and elaborate as they apply what they know to new mathematical situations.

As you plan for lessons throughout the unit, be intentional in decision-making to provide space and opportunity for all students to engage in each phase while also differentiating the mode in which students engage with each phase.

Reflections

Across the x -axis	Across the y -axis
The y -coordinate changes to its opposite . $(x, y) \rightarrow (x, -y)$	The x -coordinate changes to its opposite . $(x, y) \rightarrow (-x, y)$
across $y = x$	across $y = -x$
The coordinates switch places . $(x, y) \rightarrow (y, x)$	The coordinates switch places and become opposite . $(x, y) \rightarrow (-y, -x)$

On-Ramp

Use the videos and practice problems in the On-Ramp to Geometry personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish.

- Identifying Transformations



Section Coherence

Section 1: Sequence of Transformations (Lessons 1-5)

Benchmarks Addressed	Learning Targets	
MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates. <i>Clarification 1: Instruction includes the connection of transformations to functions that take points in the plane as inputs and give other points in the plane as outputs.</i> <i>Clarification 2: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.</i> <i>Clarification 3: Within the Geometry course, rotations are limited to 90°, 180° and 270° counterclockwise or clockwise about the center of rotation, and the centers of rotations and dilations are limited to the origin or a point on the figure.</i> MA.912.GR.2.2 Identify transformations that do or do not preserve distance. <i>Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or coordinates.</i> <i>Clarification 2: Instruction includes recognizing that these transformations preserve angle measure.</i> MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure. <i>Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.</i> <i>Clarification 2: Within the Geometry course, figures are limited to triangles and quadrilaterals and rotations are limited to 90°, 180° and 270° counterclockwise or clockwise about the center of rotation.</i> <i>Clarification 3: Instruction includes the understanding that when a figure is mapped onto itself using a reflection, it occurs over a line of symmetry.</i> MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane. <i>Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.</i> <i>Clarification 2: Instruction includes two or more transformations.</i>	Lesson 1	- I can relate a translation on the coordinate plane to an algebraic notation of the transformation. - I can relate a rotation about the origin on the coordinate plane to an algebraic notation of the transformation.
	Lesson 2	I can relate a reflection on the coordinate plane to algebraic notation of the transformation.
	Lesson 3	- I can identify transformations that preserve distance. - I can specify a sequence of transformations that will map a given figure onto another congruent figure.
	Lesson 4	- I can draw an image given a preimage and a written description of a sequence of transformations on a coordinate plane. - I can describe a transformation on the coordinate plane using coordinate notation with algebraic descriptors.
	Lesson 5	- I can draw an image given a preimage and an algebraic description of a sequence of transformations on a coordinate plane. - I can write a transformation on the coordinate plane using coordinate notation with algebraic descriptors.

Section 2: Geometric Congruence (Lessons 6-8)

Benchmarks Addressed	Learning Targets	
MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure. <i>Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.</i> <i>Clarification 2: Within the Geometry course, figures are limited to triangles and quadrilaterals and rotations are limited to 90°, 180° and 270° counterclockwise or clockwise about the center of rotation.</i> <i>Clarification 3: Instruction includes the understanding that when a figure is mapped onto itself using a reflection, it occurs over a line of symmetry.</i>	Lesson 6	- I can describe a sequence of transformations that map one figure onto another congruent figure. - I can draw an image given a preimage and a sequence of transformations on a coordinate plane.
	Lesson 7	- I can describe a sequence of transformations that will map a preimage and an image to show congruence. - I can describe a sequence of transformations that map one figure onto another congruent figure.
	Lesson 8	- I can determine a sequence of transformations that map one figure onto another congruent figure. - I can match the written and algebraic descriptions of a sequence of transformations given the preimage and image of a figure.
MA.912.GR.2.6 Apply rigid transformation to map one figure onto another to justify that the two figures are congruent. <i>Clarification 1: Instruction includes showing that the corresponding sides and the corresponding angles are congruent.</i>		



Vertical Alignment

Grade 8 Unit 9
Transformations



Geometry Unit 4
Rigid Motions
and Congruence

Prior Course(s)

MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.

MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.

MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.

MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.

Course Level

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

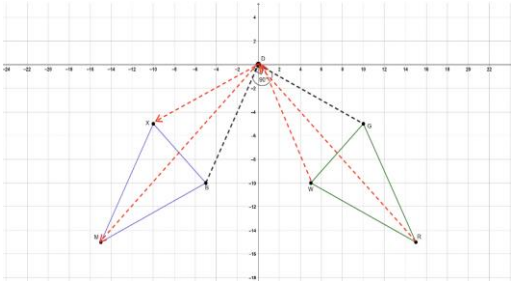
MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.

MA.912.GR.2.6 Apply rigid transformation to map one figure onto another to justify that the two figures are congruent.

4.1 Algebraic Descriptions for Translations and Rotations

Lesson Introduction

 Benchmark	MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.		 Learning Targets	- I can relate a translation on the coordinate plane to an algebraic notation of the transformation. - I can relate a rotation about the origin on the coordinate plane to an algebraic notation of the transformation.	
 Benchmark Coherence	Building On MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.		Working Toward N/A		
 Essential Questions	- How can translations be represented algebraically? - What is the same or different about horizontal and vertical translations? - How can rotations be represented algebraically? - How are rotating clockwise and counterclockwise similar? How are they different?				
 Lesson Narrative	Students collaborate while making connections using multiple representations of translations and rotations. Students make use of tables to represent values as coordinate pairs to understand how to write translations and rotations using algebraic notation. The majority of this lesson is collaborative in nature, with discussions serving as the primary driver of learning throughout the 4.1.2 and 4.1.3 Collaboration components that act as the bulk of the lesson.				
 Component Summary	4.1.1 Warm-Up (~5 min.)	Students learn about pharmacists as a Career Profile (video) (<i>SE p. 177</i>)	4.1.3 Collaboration (15 min.)	Students explore the relationship between image rotations and algebraic representation or notation (<i>SE p. 180</i>)	
	4.1.2 Collaboration (15 min.)	Students explore the relationship between image translations and algebraic representation or notation (<i>SE p. 178</i>)	4.1.4 Bring It Together (10 min.)	Students summarize and synthesize learning targets, including abstract algebraic notation (<i>SE p. 183</i>)	
	4.1.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 183</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Translation - Rotation <u>Additional Terms:</u> N/A		(Optional) Chart paper (Optional) Markers		4.1.1 Warm-Up contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse clockwise and counterclockwise. - Students may confuse which images are considered preimage and what “prime notation” means in context. - Students might confuse x- and y-values in the coordinate plane.	- The connections between the images, the tables, and the algebraic descriptions are important in developing an understanding of transformations. Allow the students to grapple with writing the conjectures so those connections are made. - Consider common misconceptions from Unit 1 and student understanding shown in the unit to help determine collaboration group size and makeup.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 4.1.2 Collaboration positions students to notice the effects of rotation on vertex coordinates. Consider having students apply Notice and Wonder routines within their collaboration group to help each member of the group discover the relationship between changing values of coordinates and the algebraic representations of rotations.	MA.K12.MTR.2.1: Multiple Representations Students will be working to represent translations and rotations algebraically. Clarifications within this standard emphasize connections between concept and representation. Utilize the progression of the lesson to highlight the process by which students are moving from image to abstract using algebraic notation.
 Differentiate Instruction	For a kinesthetic entry point, consider using physical manipulatives such as patty paper or cutouts for students to explore each of the transformations.	
	For example, students can use patty paper for 4.1.3 Collaboration as they map the transformation for the triangle GWR .	
		
Lesson Notes:		

4.1.1 Warm-Up: A Career in Pharmacy

Component Overview: Display the “Career Profile – Pharmacy” video for the entire class. After students watch this two-minute video, prompt the students to read and answer the questions in the Warm-Up independently.



4.1.1 Warm-Up: A Career in Pharmacy

A career as a pharmacist is just one of many jobs in the health care field. Pharmacists need an understanding of both science and mathematics. Pharmacists also need to work with patients on their prescriptions to ensure the medication is working as it should.



1. What is one thing you learned about pharmacists that you did not know?

Answers will vary.

2. How does paying attention to detail help you to learn mathematics?

Answers will vary. Sample Answer: Paying attention to details when working a problem will ensure that no steps are missed, and each step is done correctly.



Consider positioning students to discuss question prompts rather than writing their answers. This entry point might engage auditory learners. Use sentence stems to encourage conversation such as:

- “I had to pay attention when problem-solving when...”
- “One detail that I have to remember a lot in math class is...”



Do you know someone who is a pharmacist or works in pharmaceuticals? Consider bringing in a visitor for your class to ask questions and learn from. Additionally, consider having students return to this lesson later in the year for project-based learning around this Career Profile.

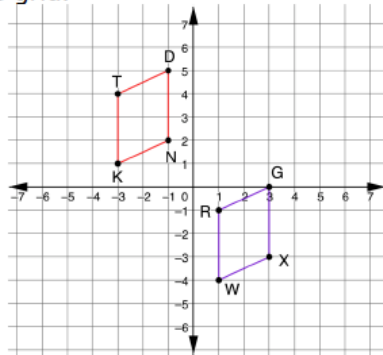
4.1.2 Collaboration: Discovering Algebraic Descriptions for Translations

Component Overview: Students will be collaborating on a series of preimages and images. The exploration stage is the same for each set of images, with students first making observations before representing the transformations algebraically. The first set of images explores the process in-depth with collaborative discussion prompts, while the remaining sets of shapes provide an exploration.



4.1.2 Collaboration: Discovering Algebraic Descriptions for Translations

1. Parallelograms $DNKT$ and GXR are shown on the coordinate grid.



- Describe the transformation if parallelogram $DNKT$ is the preimage.
DNKT is shifted to the right 4 then shifted down 5 OR DNKT is shifted down 5 then shifted to the right 4.
- Describe the transformation if parallelogram GXR is the preimage.
GXR is shifted to the left 4 then shifted up 5 OR GXR is shifted up 5 then shifted to the left 4.
- Complete the table by writing the x -value of each vertex in the columns provided. Then, determine the number of units that should be added or subtracted to the x -value for the vertices of parallelogram $DNKT$ to get the x -value for the vertices of parallelogram GXR .

Vertex	x	Vertex	x	Description
D	-1	G	3	Add 4
N	-1	X	3	Add 4
K	-3	W	1	Add 4
T	-3	R	1	Add 4



Consider highlighting student responses that differ by which parallelogram is identified as the preimage. This discourse can position students to better understand why prime notation is important in mapping translations.



Consider prompting questions for students working in pairs who are stuck.
 How are the answers to questions 1a and 1b the same?
 How are they different?
 What does that make you think about describing transformations?
 Why does it matter which parallelogram is the preimage?



As students work in pairs, consider assigning students to image and preimage or x -values and y -values. Charge students to compare and contrast their findings with their partner and make conjectures to answer question 1e based on those findings.

4.1.2 Collaboration: Discovering Algebraic Descriptions for Translations (continued)

- d. Complete the table by writing the y -value of each vertex in the columns provided.

Then, determine the number of units that should be added or subtracted to the y -value for the vertices of parallelogram $DNKT$ to get the y -value for the vertices of parallelogram $GXWR$.

Vertex	y	Vertex	y	Description
D	5	G	0	Subtract 5
N	2	X	-3	Subtract 5
K	1	W	-4	Subtract 5
T	4	R	-1	Subtract 5

Transformations can be described using words, as well as using a notation.

- e. Use the tables to complete the algebraic description of the transformation if parallelogram $DNKT$ is the preimage.

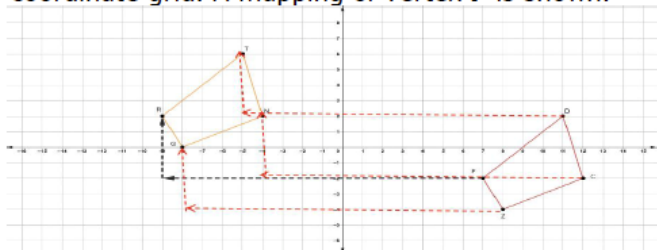
$$(x, y) \rightarrow (x + 4, y - 5)$$

- f. Discuss with your partner how the algebraic description of the transformation would change if parallelogram $GXWR$ is the preimage.

Sample response: The opposite movement is being made and that creates a situation that uses opposite operations.

$$(x, y) \rightarrow (x - 4, y + 5)$$

2. Quadrilaterals $GRTN$ and $ZFDC$ are shown on the coordinate grid. A mapping of vertex F is shown.



- a. Map the transformation for the other vertices of quadrilateral $ZFDC$.



Use prompting questions for students who are struggling to make the connection between the translations and the changes in values.

How does the description of the operation for the x -value relate to the translation?

How does the description of the operation for the y -value relate to the translation?



Prompt students to make connections from question 1f to question 1a and 1b. Students who may not have understood the importance of the order may understand that the algebraic description is incorrect if the preimage and image are switched.

4.1.2 Collaboration: Discovering Algebraic Descriptions for Translations (continued)

- b. Discuss with your partner how the algebraic description of the transformation, $(x, y) \rightarrow (x - 16, y + 4)$, is revealed in the mapping. Write a summary of your discussion.
Answers will vary. Sample answer: ZFDC is shifted to the left 16 units then shifted up 4 units.

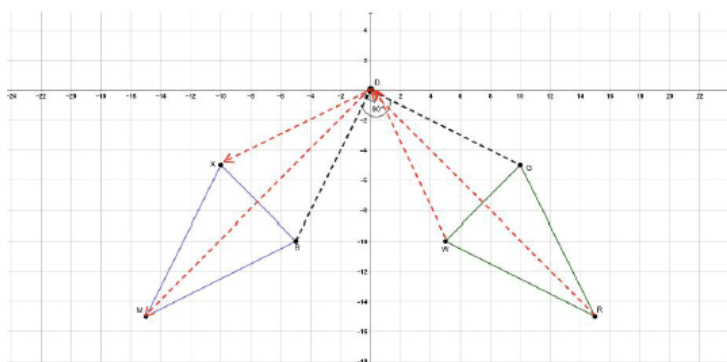
4.1.3 Collaboration: Discovering Algebraic Descriptions for Rotations

Component Overview: Students will be collaborating on a series of preimages and images. The exploration stage is the same for each set of images, with students first making observations before representing these movements algebraically. The first set of images explores the process in-depth with collaborative discussion prompts, while the remaining sets of shapes provide an exploration.



4.1.3 Collaboration: Discovering Algebraic Descriptions for Rotations

1. Triangles BXM and GWR are shown on the coordinate grid, where $\triangle GWR$ is the preimage. A mapping of vertex G is shown.



- a. Map the transformation for the other vertices of triangle GWR .
- b. Describe the transformation if triangle GWR is the preimage.
 GWR rotated 90° clockwise about point D
- c. Discuss with your partner how to use the table of values to describe the relationship of the corresponding vertices using an algebraic description.

Triangle GWR

Vertex	x	y
G	10	-5
W	5	-10
R	15	-15

Triangle BXM

Vertex	x	y
B	-5	-10
X	-10	-5
M	-15	-15



Consider providing resources like patty paper or a protractor for students to identify the degree of rotation as 90° as well as mapping the vertices to visually represent the right angle (as seen in the answer key).



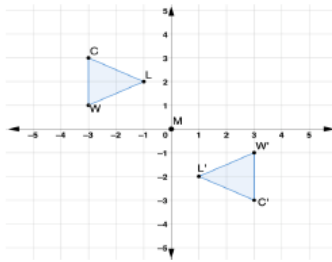
Consider revisiting question 1b, highlighting the direction of the rotation (clockwise), when students are answering question 3b later in the activity.

4.1.3 Collaboration: Discovering Algebraic Descriptions for Rotations (continued)

- d. Write a conjecture that you and your partner think will work for the algebraic description of the rotation.
 $(x, y) \rightarrow (y, -x)$

Answers will vary. Sample Answer: Change the sign of the x -value by multiplying by -1 . Then, swap the x - and y -values.

2. Triangles CLW and $C'L'W'$ are shown on the coordinate grid. Triangle CLW was rotated 180° about M to create $\triangle C'L'W'$.



Triangle CLW

Triangle $C'L'W'$

Vertex	x	y
C	-3	3
L	-1	2
W	-3	1

Vertex	x	y
C'	3	-3
L'	1	-2
W'	3	-1

- a. It was stated that $\triangle CLW$ was rotated 180° about M to create $\triangle C'L'W'$. Was that rotation clockwise or counterclockwise?

Answers will vary. It could be either clockwise or counterclockwise.



Notice and Wonder

Provide students with question stems using the Notice and Wonder routine to provide intentional scaffolding for making connections between the vertices, rotations, and algebraic descriptions.

Notice: What do you notice about the values in Vertex G compared to Vertex B ?

Wonder: Does this also hold true for Vertex W and Vertex X ? Vertex R and Vertex M ?

Question 1d: What is your conjecture for how to represent this algebraically?

This line of question stems can apply to questions 2 and 3 as well as for when students collaborate and explore in pairs.



What do you notice about a clockwise rotation of 180° and a counterclockwise rotation of 180° ?

What other transformations could be used to produce this same image from this preimage? (Example: a reflection of the y -axis followed by a reflection over the x -axis.)

4.1.3 Collaboration: Discovering Algebraic Descriptions for Rotations (continued)

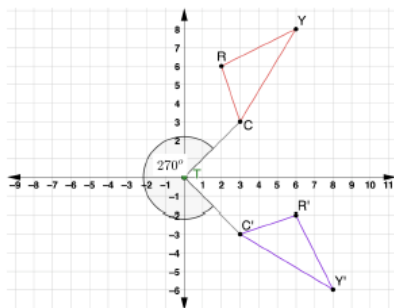
- b. Discuss with your partner how your previous conjecture works with rotating $\triangle CLW$ 180° about M . If necessary, write a new conjecture that works for rotations of 180° about the origin. Include the direction of the rotation determined. Write a summary of your discussion.

Answers will vary. Sample Answer: A rotation of 180° is the same as two rotations of 90° . For a 90° clockwise rotation, you multiply the x -value by -1 then swap the x - and y -values. This is represented by the algebraic description is $(y, -x)$. Rotating 90° clockwise again you will multiply the new x -value, which is y , by -1 and swap the new x - and y -values again. This is represented by the algebraic description of $(-x, -y)$. Therefore the algebraic description for a 180° clockwise rotation will produce an algebraic description of $(-x, -y)$.

- c. Assuming $\triangle C'L'W'$ was the preimage and is rotated 180° about the origin, test whether your previous or new conjecture works to describe this rotation. Include the direction of rotation.

Answers will vary. Sample Answer: Yes, the conjecture works moving in either direction, clockwise or counterclockwise because of the 180° movement.

3. Triangles YCR and $Y'C'R'$ are shown on the coordinate grid. $\triangle YCR$ was rotated 270° counterclockwise about T to create $\triangle Y'C'R'$.



For additional complexity, have students consider rotations around a point other than the origin.



For questions 3-5, consider alternative ways for expressing terms like “clockwise” and “counterclockwise.” Students growing up with digital clocks may be unfamiliar with those terms.



Monitor for students who may be struggling with summarizing their discussion. The student is not expected to express the relationship using algebraic descriptions yet. If students can visually relate the transformation, then build from the visual to the algebraic. If students can relate the transformation algebraically but cannot see it visually, then help them notice the angle through the mapping and use of a protractor.



If a student chose counterclockwise for Part a, argument here still applies. The only difference is that rotating point $(-y, x)$ counterclockwise 90° twice still produces description $(-y, -x)$.

4.1.3 Collaboration: Discovering Algebraic Descriptions for Rotations (continued)

Triangle YCR

Vertex	x	y
Y	6	8
C	3	3
R	2	6

Triangle $Y'C'R'$

Vertex	x	y
Y'	8	-6
C'	3	-3
R'	6	-2



- a. Discuss with your partner how the ordered pairs for $\triangle YCR$ and $\triangle Y'C'R'$ could be used to write a conjecture that works for rotations of 270° counterclockwise about the origin. Write a summary of your discussion.

$$(x, y) \rightarrow (y, -x)$$

Answers will vary. Sample answer: the x -value is multiplied by -1 , then the x - and y - value switch.

- b. Test your conjecture for rotations of 270° counterclockwise by looking at $\triangle BXM$ and $\triangle GWR$ from question 1. Assume $\triangle GWR$ is the preimage and was rotated 270° counterclockwise about the origin. If necessary, adjust your conjecture.

- c. Discuss with your partner why the same conjecture should work for a rotation of 270° counterclockwise and 90° clockwise about the origin.

Answers will vary. Sample answer: both rotations result in the same location on the coordinate plane.

If students struggle responding to this prompt, have them compare this transformation to the first rotation. How are they the same? How are they different?

Question 3b intentionally does not give direction so that students include direction in their verification and discussion of their conjecture.

By question 3c, students should be able to draw the conclusion that 270° clockwise is an equivalent rotation to 90° counterclockwise and that 270° counterclockwise is an equivalent rotation to 90° clockwise.

4.1.4 Bring It Together: Algebraic Descriptions for Translations and Rotations

Component Overview: Students should synthesize their findings from the 4.1.2 and 4.1.3 Collaborations by summarizing algebraic descriptions for types of translations and rotations.



4.1.4 Bring It Together: Algebraic Descriptions for Translations and Rotations

Translations	
Right	Left
Add to the x -coordinate $(x, y) \rightarrow (x + a, y)$	Subtract from the x -coordinate $(x, y) \rightarrow (x - a, y)$
Up	Down
Add to the y -coordinate $(x, y) \rightarrow (x, y + b)$	Subtract from the y -coordinate $(x, y) \rightarrow (x, y - b)$

Rotations clockwise about the origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
$(x, y) \rightarrow (y, -x)$	$(x, y) \rightarrow (-x, -y)$	$(x, y) \rightarrow (-y, x)$
Rotations counterclockwise about the origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
$(x, y) \rightarrow (-y, x)$	$(x, y) \rightarrow (-x, -y)$	$(x, y) \rightarrow (y, -x)$

1. What do you notice about the algebraic descriptions for rotations?

Answers will vary. Rotations of 90° clockwise are equivalent to rotations of 270° counterclockwise. Rotations of 270° clockwise are equivalent to rotations of 90° counterclockwise. A rotation of 180° has the same result no matter the direction of the rotation.



Use this section to facilitate synthesis, not memorization. Consider using this section as notetaking for students during the Collaboration activities.



Consider a foldable, Frayer Model, or type of interactive notebook for this summary. Examples, nonexamples, and verbal descriptions will provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.

4.1.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



4.1.5 Wrap-Up: Cool Down

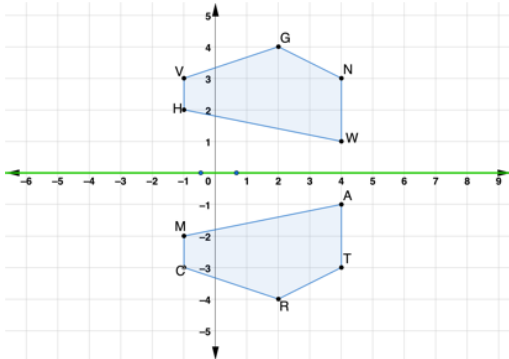
1. A student asks you for help with algebraic descriptions for rotations because they are stuck on how to do it. Describe the process using what you learned in this lesson, highlighting any potential misconceptions they may have.

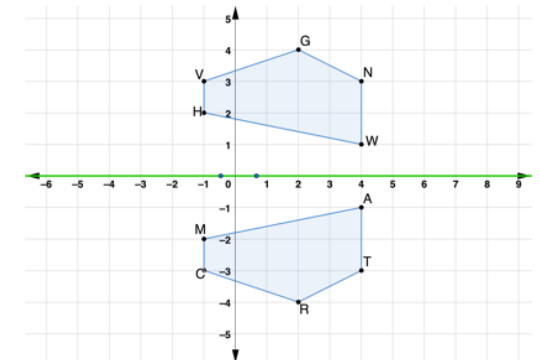
Answers will vary.

4.2 Algebraic Descriptions for Reflections

Lesson Introduction

 Benchmark	MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.		 Learning Target	I can relate a reflection on the coordinate plane to an algebraic notation of the transformation.
 Benchmark Coherence	Building On MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.		Working Toward N/A	
 Essential Questions	- What is a reflection? - How can reflections be represented algebraically?			
 Lesson Narrative	The main part of the lesson, 4.2.2 Exploration, immerses students in an in-depth exploratory discovery of the effects of reflections across a variety of lines. Students focus on the ordered pairs of the vertices and the midpoint of line segments that connect corresponding vertices across the line of reflection. Students engage in language routines and math thinking and reasoning to draw conclusions leading toward generating algebraic descriptions of reflections.			
 Component Summary	4.2.1 Warm-Up (~5 min.)	Critical thinking problem (SE p. 184)	4.2.3 Bring It Together (5-10 min.)	Students summarize and synthesize their findings from the Exploration (SE p. 189)
	4.2.2 Exploration (25-30 min.)	Exploration of reflections across a variety of lines of reflection (SE p. 185)	4.2.4 Wrap-Up (~5 min.)	Students relate elements of reflections to a squish painting (SE p. 189)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Integer chips or other chips for 4.2.1 Warm-Up (Optional) Squish painting	N/A

 Teacher Tips	Common Misconceptions - Students may struggle differentiating between axes, values in an ordered pair, and their opposites. - Students who struggle in spatial reasoning might find it difficult to navigate the visual elements of reflections.	Things to Consider Consider using sentence stems to actively convert student verbal descriptions into mathematical language. For example, when students are describing the effects of reflection like, “They are in the same place,” provide scaffolded phrases like, “Vertex __ is the same distance away from __ as Vertex __.”
	Literacy and Language Routines Notice and Wonder 4.2.2 Exploration questions 1c-f provide two opportunities for students to make sense of their observations in mathematically meaningful ways. Encourage students to utilize Notice and Wonder techniques for each element to help each other discover and produce algebraic descriptions based on the Exploration.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection 4.2.2 Exploration question 2b provides an opportunity for students to engage in mathematical discussion by analyzing and critiquing the thinking of others. Use this as an opportunity to provide students with practice analyzing the thinking of others and then articulating their own thinking in response.
	 Differentiate Instruction	For a kinesthetic entry point, consider using physical manipulatives such as patty paper or cut-outs for students to explore the reflections in 4.2.2 Exploration. 
Lesson Notes:		



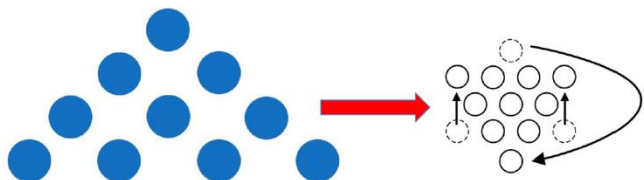
4.2.1 Warm-Up: Moving Circles

Component Overview: Students engage with a puzzle to determine how to solve it by moving only three circles.



4.2.1 Warm-Up: Moving Circles

1. Move any three circles so that the pyramid points down.



The arrows provided here are for answer key purposes. Students will only have the blue dots visible.

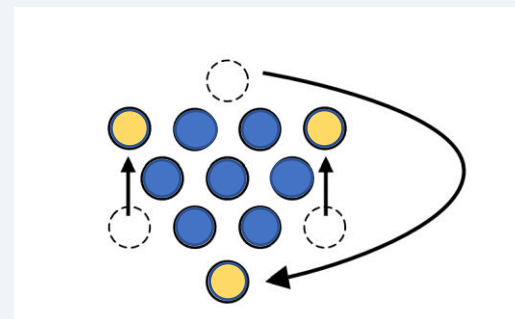


Consider prompting questions to target specific actions of Growth Mindset.

- Was the puzzle a struggle, or was it easy for you?
- What did you look for first when making pyramid point down?
- What did you try first? What strategy or different way of looking at the pyramid helped you know what to do next?



Do you have integer chips or similar manipulatives? Students could interact with this pyramid prompt for a kinesthetic learning experience. This could also be done with digital manipulatives.



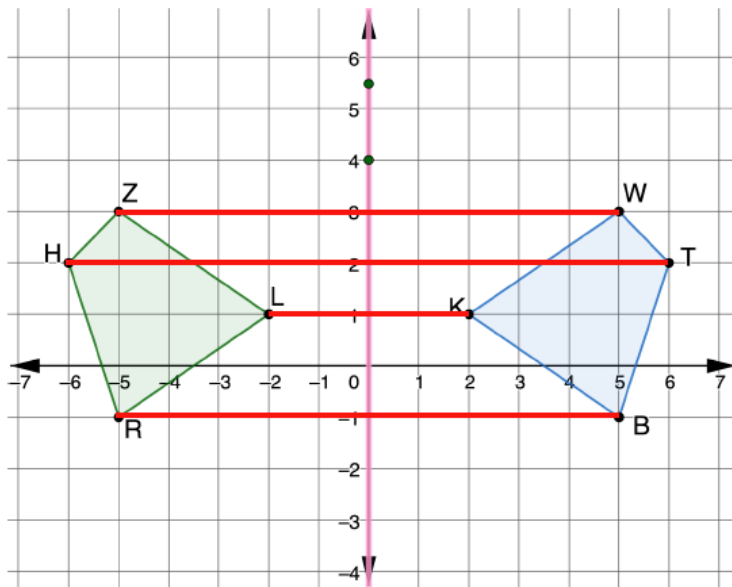
4.2.2 Exploration: Discovering Algebraic Descriptions for Reflections

Component Overview: Students will explore how changes in x -values and y -values of ordered pairs of corresponding vertices after a reflection can be used to determine algebraic descriptions of the reflections. Students explore reflections across the y -axis, x -axis, the origin, the line $y = x$, and the line $y = -x$.



4.2.2 Exploration: Discovering Algebraic Descriptions for Reflections

1. Quadrilaterals $HZLR$ and $TWKB$ are shown on the coordinate grid. Quadrilateral $HZLR$ was reflected across the y -axis to create quadrilateral $TWKB$.



- a. Draw a line segment between each set of corresponding vertices.
Shown on figure.
- b. Describe how the line segments between the vertices are related to the y -axis.
Answers will vary. Each vertex is the same distance from the y -axis as its corresponding vertex. The y -axis bisects each line segment. The line segments between the vertices are perpendicular to the y -axis.



This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the exploration as they engage with each new reflection. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group Bring It Together. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.



During the exploration, allow students to describe relationships using their own words first. Students informally experienced reflections in Unit 1, so the exploration helps them recall that information. For students who are struggling with descriptions, refer them to Unit 1. As you circulate, if student descriptions show misunderstanding, then ask probing questions like:

- What is the relationship $\overline{ZW}$ has with the y -axis?
- Is that relationship the same for the other line segments?

4.2.2 Exploration: Discovering Algebraic Descriptions for Reflections (continued)

c. Complete the table.

	<i>H to T</i>	<i>Z to W</i>	<i>L to K</i>	<i>R to B</i>
Distance	12	10	4	10
Ordered Pair of the Midpoint	(0,2)	(0,3)	(0,1)	(0,-1)

d. What do you notice about the midpoints?

Answers will vary. All x -values are zero. All of the midpoints are on the y -axis.

e. The table shows the coordinates of each vertex.

Vertex	x	y	Vertex	x	y
<i>H</i>	-6	2	<i>T</i>	6	2
<i>Z</i>	-5	3	<i>W</i>	5	3
<i>L</i>	-2	1	<i>K</i>	2	1
<i>R</i>	-5	-1	<i>B</i>	5	-1

Work with your partner to describe how the coordinates of the pre-image, quadrilateral $HZLR$, are related to the coordinates of the image, quadrilateral $TWKB$.

Answers will vary. The x -values are opposites and the y -values are the same.

f. Write a conjecture that you and your partner think will work for the algebraic description of a reflection.

Answers will vary. Since the polygon is reflecting over the y -axis, the x -values change to their opposite sign.

$(x, y) \rightarrow (-x, y)$

g. Discuss with your partner if the algebraic description will work for any reflection. Summarize your conclusion.

Answers will vary. The point or line of reflection matters. Reflections over the y -axis will follow that description, but if the line of reflection is not the y -axis, then a new algebraic description would be needed.



Notice and Wonder Routine

Notice the intentional scaffolding of language routine elements embedded within the Exploration. Questions 1c-1d and 1e-1f provide the space and opportunity for students to discover how the ordered pairs of the vertices after a reflection can be used to generate algebraic descriptions. Students may not be engaging in a formal Notice and Wonder routine, but they can be prompted to notice and wonder about what they see for additional entry points when stuck in parts like question 1e.

Emphasize the word “opposite” in student discussion, as this word is seen in later questions of this lesson.

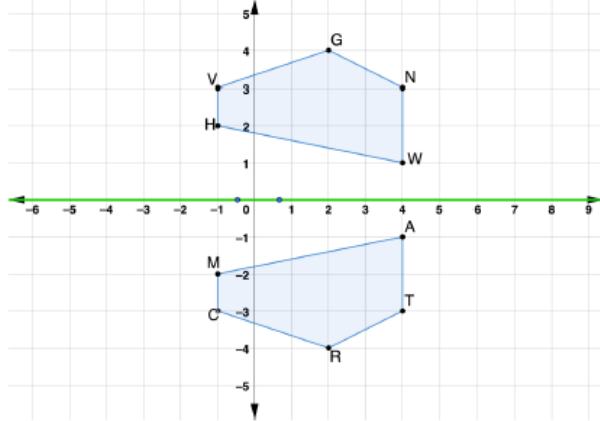


MA.K12.MTR.5.1: Patterns and Structure

The exploration provides students the opportunity to use patterns and structure to help understand and connect previously learned concepts (reflections) to new concepts (representing them with algebraic descriptions).

4.2.2 Exploration: Discovering Algebraic Descriptions for Reflections (continued)

2. Polygons $MCRTA$ and $HVGNW$ are shown on the coordinate grid. Polygon $MCRTA$ was reflected across the x -axis to create polygon $HVGNW$.



- a. Describe the relationship that the line segments between each set of corresponding vertices have with the x -axis.

Answers will vary. Each vertex is the same distance from the x -axis as its corresponding vertex. The line segments between corresponding vertices are perpendicular to the x -axis.

- b. Wendall and Samah were working together on writing an algebraic description for reflections across the x -axis. Each student's algebraic description and explanation of their thinking is shown in the table.

Wendall	Samah
$(x, y) \rightarrow (-x, y)$	$(x, y) \rightarrow (x, -y)$
Because the polygon flipped over the x -axis, the x becomes its opposite.	Because above the x -axis all y -values are positive and below the x -axis all y -values are negative, the y becomes its opposite.



Monitor student discussion for question 2a. If necessary, prompt students to use the scaffolding from question 1 similarly in question 2. Draw line segments between each set of corresponding vertices from polygon $MCRTA$ to polygon $HVGNW$.

Prompting questions could include the following:

- What do you notice about the midpoints of the line segments?
- What do you notice about the coordinates of each ordered pair?
- What do you notice about the line of reflection and the line segments?
- How are these relationships similar to and different from the previous reflection?



MA.K12.MTR.4.1: Discussion and Reflection

Students have the opportunity to critique the reasoning and thinking of others. Consider having students or their partners read this passage out loud for others to listen and respond to in conversation. The error analysis of question 2b has students look at the common misconception of a reflection across the x -axis resulting in the x -value changing.

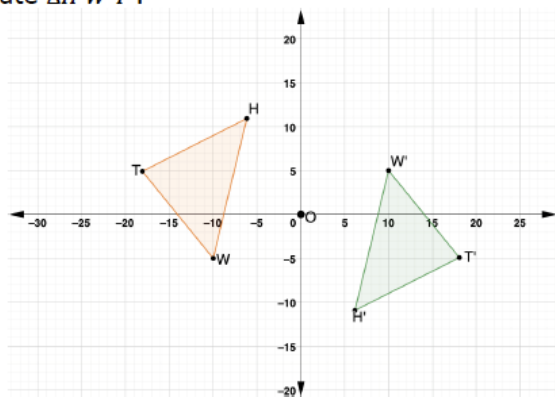
4.2.2 Exploration: Discovering Algebraic Descriptions for Reflections (continued)

Discuss with your partner which, if any, algebraic description is correct. Use at least two ordered pairs from polygons $MCRTA$ and $HVGNW$ to defend your choice. Write a summary of your discussion.

Samah is correct.

Answers will vary. Samah is correct because $A(4, -1)$ becomes $W(4, 1)$ and $R(2, -4)$ becomes $G(2, 4)$. You can see that the x -values remained the same but the y -values are now the opposite sign.

3. Triangles HWT and $H'W'T'$ are shown on the coordinate grid. Triangle HWT was reflected across the origin to create $\triangle H'W'T'$.



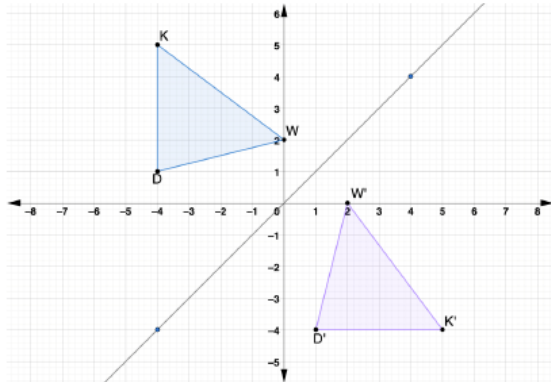
- Discuss with your partner the algebraic description for a reflection across the origin.
 $(x, y) \rightarrow (-x, -y)$
- Re-examine the algebraic descriptions of rotations. Which rotation is similar to a reflection across the origin?
 180° rotation



Question 3b seeks to build a conceptual connection to the previous lesson. Consider how to best engage students with the two elements of question 3 by beginning with a question like, “Does this look familiar? Why or why not?” and suggest they start with question 3b before finishing on question 3a.

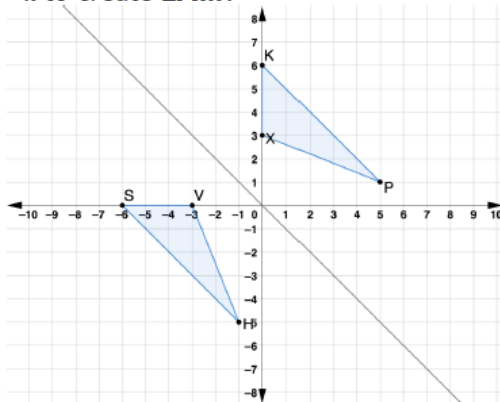
4.2.2 Exploration: Discovering Algebraic Descriptions for Reflections (continued)

4. Triangles KWD and $K'W'D'$ are shown on the coordinate grid. Triangle KWD was reflected across the line $y = x$ to create $\triangle K'W'D'$.



Use the coordinates to determine the algebraic description for a reflection across the line $y = x$.
 $(x, y) \rightarrow (y, x)$

5. Triangles HSV and PKX are shown on the coordinate grid. Triangle HSV was reflected across the line $y = -x$ to create $\triangle PKX$.



Use the coordinates to determine the algebraic description for a reflection across the line $y = -x$.
 $(x, y) \rightarrow (-y, -x)$



If time permits, transform your classroom into a coordinate plane and challenge students to create a mathematically accurate reflection using elements like desks, chairs, bags, and more!

4.2.3 Bring It Together: Algebraic Descriptions for Reflections

Component Overview: Students summarize and synthesize what they have discovered about reflections across both axes, $y = x$, and $y = -x$, including representing each by its algebraic description.



4.2.3 Bring It Together: Algebraic Descriptions for Reflections

Reflections	
Across the x -axis	Across the y -axis
The y -coordinate changes to its <u>opposite</u> . $(x, y) \rightarrow (x, -y)$	The x -coordinate changes to its <u>opposite</u> . $(x, y) \rightarrow (-x, y)$
Across $y = x$	Across $y = -x$
The coordinates <u>switch places</u> . $(x, y) \rightarrow (y, x)$	The coordinates <u>switch places</u> and become <u>opposite</u> . $(x, y) \rightarrow (-y, -x)$



Use this section to encourage synthesis, not memorization. Call on students to share their ideas following the exploration to highlight student voice and solidify takeaways from the Exploration.

4.2.4 Wrap-Up: Cool Down

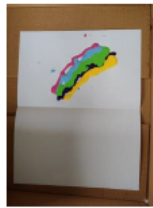
Component Overview: Students apply mathematical thinking and reasoning learned throughout the lesson to connect a line of reflection to art.



4.2.4 Wrap-Up: Cool Down

A reflection can be thought of as folding or "flipping" an object over the line of reflection. A squish painting is a fun application of a line of reflection.

To make a squish painting, a large sheet of construction paper is folded in half then opened up. Paint is dotted along one side of the fold. The paper is folded, and the folded paper is rubbed, pounded, and squished to move the paint around. The paper is then opened up to reveal the squish painting.



1. How does a squish painting show that in a reflection, each point of the original figure and its image are the same distance away from the line of reflection?

Answers will vary. In a squish painting, you can see how close each part of the painting is to the crease in the paper. You can see that the picture is a mirror image. The fold is the line of reflection.

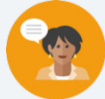


Consider what this Cool Down might look like as a performance task or interdisciplinary project with an art class. Students could be immersed into an exploration of reflections in art to establish a strong foundation for key elements of reflections, while the 4.2.2 Exploration can serve to quantify and make mathematical sense of their findings of reflections in art.

4.3 Sequences of Transformations

Lesson Introduction

 Benchmarks	MA.912.GR.2.2 Identify transformations that do or do not preserve distance.		 Learning Targets	- I can identify transformations that preserve distance. - I can specify a sequence of transformations that will map a given figure onto another congruent figure.
 Benchmark Coherence	Building On			
	MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.		N/A	
 Essential Questions	- How can you determine if a transformation preserves distance? - How can you use a sequence of transformations to map a given figure onto another congruent figure?			
 Lesson Narrative	In the lesson, students will apply what they have learned about translations, rotations, and reflections in the prior two lessons to begin piecing them together in a sequence of transformations.			
 Component Summary	4.3.1 Warm-Up (~5 min)	Fill-in-the-blank statements (<i>SE p. 190</i>)	4.3.4 Your Turn! (10 min)	Independent practice on describing transformations and writing transformations using algebraic descriptions (<i>SE p. 194</i>)
	4.3.2 Guided Instruction (15-20 min)	Guided instruction on how to sequence multiple transformations (<i>SE p. 191</i>)		
		4.3.3 Guided Practice (10 min)	Completing a table with the transformation and algebraic description for a specified mapping (<i>SE p. 194</i>)	4.3.5 Wrap-Up (~5 min)
 Resources	Glossary		Materials	
	Key Terms: N/A		Patty paper	
	Additional Terms: - Origin x-axis y-axis			
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions - Students may apply the wrong sign when completing a reflection. - Students may write the wrong algebraic description for a different rotation type than the one intended. - Students may struggle to determine the correct sequence to complete the mapping.	Things to Consider Consider elements of instruction from Unit 1 resources that can activate prior knowledge during this lesson. Both Unit 1 and the activities in this lesson purposefully position students as the sense makers in order to make meaningful connections between their work with figures in Unit 1 and transformations in this lesson.
	Literacy and Language Routines Compare and Connect During 4.3.2 Guided Instruction activity, students receive guided instruction on the sequence of transformations. While guiding students through the activity, it will be important to share teacher thinking so that students can make the connection that the activity intends.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure Throughout the core activities of this lesson, students will be applying their prior knowledge of isolated transformations to specifying a sequence of transformations to map a given figure onto another congruent figure.
	 Differentiate Instruction Students who struggle with individual transformations will likely carry that over to sequences of transformations. In the beginning, consider breaking the sequences apart into separate transformations before combining into one complete problem. Providing this entry point will highlight student understanding of individual transformations, which can be used to celebrate what has been learned and to determine remediation steps for what still needs work.	
Lesson Notes:		

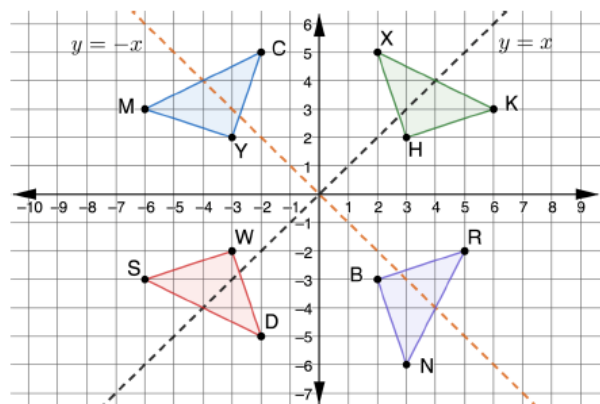
4.3.1 Warm-Up: Bell Work

Component Overview: Students will complete a fill-in-the-blank activity that assesses their prior learning with reflections.



4.3.1 Warm-Up: Bell Work

Triangles MCY , KXH , NRB , and SWD are shown on the coordinate grid.



Complete each statement.

1. When $\triangle MCY$ is reflected across **the y -axis**, it will map onto $\triangle KXH$.
2. When $\triangle NRB$ is reflected across **$y = x$** , it will map onto $\triangle MCY$.
3. When $\triangle SWD$ is reflected across the **x -axis**, it will map onto $\triangle MYC$.



This activity will provide insight into what information students have retained from the prior lessons. Watch how students perform on this Warm-Up to get an idea for which students may need more support during this lesson.



How does the line of reflection impact what happens to the image? What are some ways that this happens?

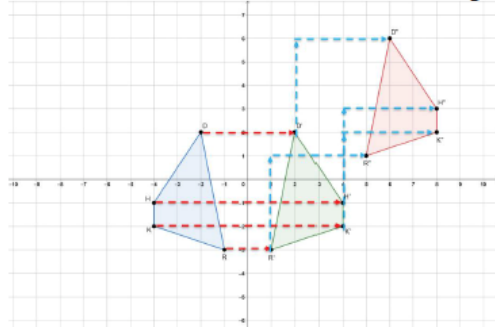
4.3.2 Guided Instruction: Sequence of Transformations

Component Overview: Students will be introduced to a sequence of transformations. This activity is designed to connect their prior learning of isolated transformations to sequences of transformations.



4.3.2 Guided Instruction: Sequence of Transformations

1. Quadrilateral $RKHD$ was transformed to create quadrilateral $R'K'H'D'$. Then, quadrilateral $R'K'H'D'$ was transformed to create quadrilateral $R''K''H''D''$. The quadrilaterals are shown on the coordinate grid.



- a. On the graph, show the mappings of the vertices of the first transformation.
Shown on graph in red.
- b. On the graph, show the mappings of the vertices of the second transformation.
Shown on graph in blue.

- c. Complete the table.

	Quadrilateral $RKHD$ Transformed to Quadrilateral $R'K'H'D'$	Quadrilateral $R'K'H'D'$ Transformed to Quadrilateral $R''K''H''D''$
Written Description	Reflection across the y -axis.	Vertical translation up 4 units. Horizontal translation right 4 units.
Algebraic Description	$(x, y) \rightarrow (-x, y)$	$(x, y) \rightarrow (x + 4, y + 4)$



If available, consider using physical manipulatives or patty paper to connect each step of the sequence for students to connect individual transformation to the overall sequence.



Pay close attention to how students write their algebraic description. Many students will be able to choose the correct reflection but may inaccurately write in the algebraic description.

Ask students to notice the direction of the vertices when naming the preimage and image for a reflection. For example, $RKHD$ is named clockwise, starting at R , while $R'K'H'D'$ is named counterclockwise, starting at R' . With a reflection, the direction of the vertices changes to the opposite direction. Relate this to the change in direction when taking a picture with a front-facing selfie camera versus the back camera.



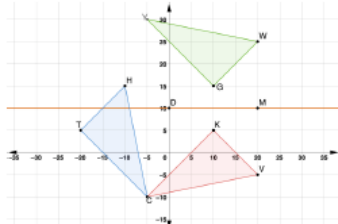
Students first encountered transformations that preserve distance in Unit 1. The figures were on the plane, not in the coordinate plane. The act of mapping the vertices could be extended to mapping all points, and because all of the points can be mapped to corresponding points, the figures are congruent.

4.3.2 Guided Instruction: Sequence of Transformations (continued)

- d. Explain how to know if a sequence of transformations preserves distance.

Answers will vary. Sample response: Each of the vertices on the quadrilaterals are the same distance apart after each transformation.

2. Triangles HTC , VKC , and WGY are shown on the coordinate grid. Triangle HTC is the pre-image. Triangle VKC is the result of the first transformation, and $\triangle WGY$ is the result of the second transformation.



- a. Describe the first transformation.
 $\triangle HTC$ rotated 90° clockwise about point C to create $\triangle VKC$.
- b. Use patty paper to show that the transformation preserved distance.
- c. Describe the second transformation.
 $\triangle VKC$ reflected over $\overline{DM}$ ($y = 10$) to create $\triangle WGY$.
- d. Use patty paper to show that the second transformation preserved distance.



Position students to make connections between their exploration of congruence in Unit 1 and their use of patty paper for mapping figures on the coordinate plane. Consider utilizing activities, videos, or visuals from Unit 1 as additional scaffolding for students to make those connections.



For question 2b, allow the students to try it on their own or with a partner. Based on how familiar your students are with patty paper, consider giving them general instructions and example of using the patty paper to make a rotation.

4.3.2 Guided Instruction: Sequence of Transformations (continued)

- e. Work with your partner to complete the statement.

Because the ☐ center of rotation ☐ ΔHTC of ☐ line of reflection ☐ ΔVKC

is ☐ $(0, 0)$, ☐ $(-5, -10)$, ☐ $y = 0$, ☐ $y = 10$, an algebraic description

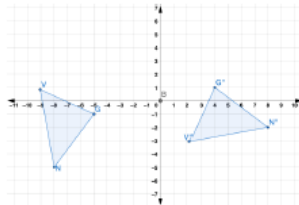
☐ can ☐ cannot be used.

- f. When Judeline and Carmen completed the statement above, they had different correct answers. Write the second correct answer.
- Option 1: center of rotation, ΔHTC , $(-5, -10)$, cannot*
- Option 2: line of reflection, ΔVKC , $y = 10$, cannot*

 To use an algebraic description of a rotation the center of rotation must be $(0, 0)$ (the origin).

 To use an algebraic description of a reflection the line of reflection must be x -axis, y -axis, $y = x$, or $y = -x$.

3. Triangles NVG and $N''V''G''$ are shown on the coordinate grid. Triangle NVG was transformed to create $\Delta N''V''G''$.



- a. Explain which features of the two triangles can be used to determine that the transformations were rigid motions.
- Answers will vary. Sample response: The lengths of the sides of the triangle remain the same.*



There are two correct answers for question 2e. The answer chosen in the first box determines the correct choice for the second and third boxes. Observe which response students choose and validate all correct answers through discussion of questions 2e and 2f. The answer to question 2f will depend on which options were chosen in question 2e. Both correct answers are shown under question 2f on the key.



- Why can't an algebraic description be used if the center of rotation $(-5, -10)$ is a point on the triangle?
- Is this a rule that would apply to other triangles as well? Why or why not?



Consider the use of an anchor chart for the blue boxes in question 2f. A T-chart could be used to illustrate algebraic description usage based on the type of transformation, including an example.



Discussion questions to consider:

- Why would the lengths of the sides of the triangles remain the same? How do you know?
- How can you determine if a rotation has occurred?
- How can you tell if the rotation was clockwise or counterclockwise?

4.3.2 Guided Instruction: Sequence of Transformations (continued)

- b. Adele told Kiriam that she knows one of the rigid motions is a rotation. Discuss with your partner which clues in the location of the vertices Adele used to draw her conclusion. Write a summary of your discussion.

Answers will vary. Sample response: the location of the vertices are rotated from the pre-image.

- c. Work with your partner to determine a sequence of transformations that includes a rotation about the point B so that when applied, $\triangle NVG$ will map onto $\triangle N''V''G''$. Complete the table.

Transformation	Mapping	Algebraic Description
<i>Sample Answer 1 Step 1: Vertical translation down 3 units and horizontal translation right 6 units.</i> <i>Sample Answer 2 Step 1: Rotation of 90° counterclockwise about point B.</i>	$\triangle NVG \rightarrow \triangle N'V'G'$	<i>Sample Answer 1 Step 1:</i> $(x, y) \rightarrow (x + 6, y - 3)$ <i>Sample Answer 2 Step 1:</i> $(x, y) \rightarrow (-y, x)$
<i>Sample Answer 1 Step 2: Rotation of 90° counterclockwise about point B.</i> <i>Sample Answer 2 Step 2: Vertical translation up 6 units and horizontal translation right 3 units.</i>	$\triangle N'V'G' \rightarrow \triangle N''V''G''$	<i>Sample Answer 1 Step 2:</i> $(x, y) \rightarrow (-y, x)$ <i>Sample Answer 2 Step 2:</i> $(x, y) \rightarrow (x + 3, y + 6)$



Consider modeling the thinking for one of the transformations and then having students determine the algebraic description. Repeat this again, but have students determine the transformation and then model the algebraic description. For the second mapping, release students to try it with a partner before reviewing the answers.

Ask students to notice the direction of the vertices when naming the preimage and image for a rotation. For example, NVG is named clockwise, starting at N , and $N'V'G'$ is also named clockwise, starting at N' . With a rotation, the direction of the vertices does not change direction.

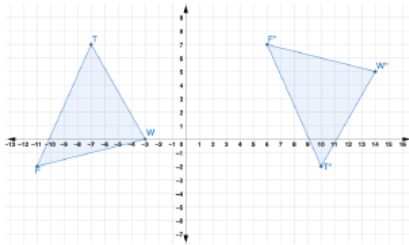
4.3.3 Guided Practice: Sequence of Transformations

Component Overview: Students work in a teacher-supported setting to complete a table with the transformation and algebraic description for a specified mapping scenario. For this activity, release students to try this question on their own or with a partner and then come back together to review as a whole group.



4.3.3 Guided Practice: Sequence of Transformations

1. Triangles TWF and $T''W''F''$ are shown on the coordinate plane.



Determine a sequence of transformations that will map $\triangle TWF$ onto $\triangle T''W''F''$. Complete the table.

Transformation	Mapping	Algebraic Description
Reflection across the x -axis.	$\triangle TWF \rightarrow \triangle T'W'F'$	$(x, y) \rightarrow (x, -y)$
Vertical translation up 5 units and horizontal translation right 17 units.	$\triangle T'W'F' \rightarrow \triangle T''W''F''$	$(x, y) \rightarrow (x + 17, y + 5)$



Consider using some of the time during this component to work with a small group of students who need additional support based on their responses and performances throughout this lesson.



For students who finish early, consider having them write an algebraic description of a transformation and then trade with a partner to write the description of the transformation. Repeat the process in the reverse order.

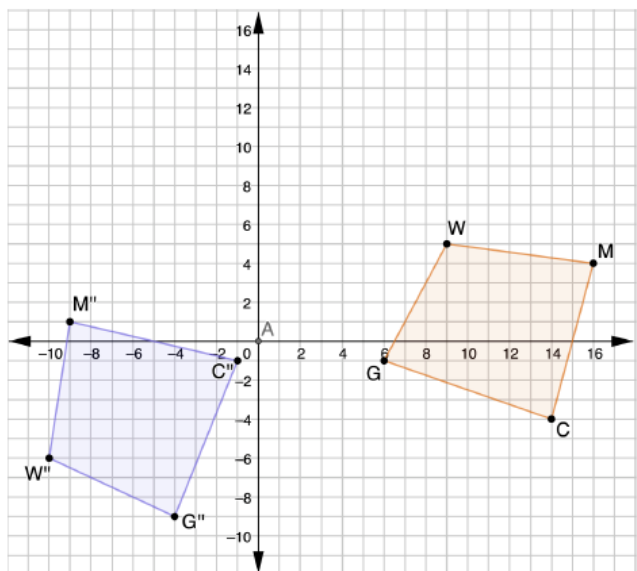
4.3.4 Your Turn!

Component Overview: Students will engage in independent practice on describing transformations and writing transformations using algebraic descriptions. For this activity, students should work independently to practice what they've learned about sequences of transformations.



4.3.4 Your Turn!

1. Quadrilateral $MCGW$ was transformed to create quadrilateral $M''C''G''W''$.



- a. Describe the sequence of transformations that will map quadrilateral $MCGW$ onto quadrilateral $M''C''G''W''$.

Answers will vary. Rotation of 90° counterclockwise about point A. Vertical translation down 15 units then horizontal translation 5 units left.

- b. Write the sequence of transformations using algebraic descriptions.

Answers will vary.

$$(x, y) \rightarrow (-y, x)$$

$$(x, y) \rightarrow (x - 5, y - 15)$$



Student performance on this task can provide insight into what students may need additional support with:

- Question 1a correct, question 1b incorrect:
 - Students can describe the correct transformation but may not understand how to accurately write in an algebraic description.
- Question 1a incorrect, question 1b correct:
 - While this is rare, it would indicate that students can represent the transformation algebraically but cannot describe the transformation.
- Question 1a incorrect, question 1b incorrect:
 - While students were unable to get the correct transformation, it is important to review whether or not their algebraic description matches their written description.

4.3.5 Wrap-Up: Reflection

Component Overview: Students will complete a self-reflection on their level of understanding of each learning target. This activity is designed to be done independently to gather data on students' self-assessments of their level of understanding from this lesson.



4.3.5 Wrap-Up: Reflection

1. Check the box that describes your understanding of each learning target.

- ☐ I can identify transformations that preserve distance and I am ready for the next lesson.

- ☐ I need some help identifying transformations that preserve distance. One area in which I need help is

_____.

- ☐ I need more support in knowing how to identify transformations that preserve distance. I am concerned I will not be ready to move on. One area in which I need help is

_____.

- ☐ I can specify a sequence of transformation that will map a given figure onto another congruent figure.

- ☐ I need some help specifying a sequence of transformation that will map a given figure onto another congruent figure. One area in which I need help is

_____.

- ☐ I need more support in knowing how to identify transformations that preserve distance. I am concerned I will not be ready to move on. One area in which I need help is

_____.

Answers will vary.



Notice the open-ended responses for the second and third options for each question. If students struggle to identify one area, consider having them choose a problem or an activity from the lesson in this blank.

4.4 Written Descriptions of a Sequence of Transformations

Lesson Introduction

 Benchmarks	MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates. MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.		 Learning Targets	- I can draw an image given a preimage and a written description of a sequence of transformations on a coordinate plane. - I can describe a transformation on the coordinate plane using written descriptions.
 Benchmark Coherence	Building On MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.		Working Toward N/A	
 Essential Questions	- How can I use the information from a written description of a series of transformations to draw an image? - How can I apply transformations to a table of values to help me draw an image?			
 Lesson Narrative	In the lesson, students will continue their work with sequences of transformations but will be transitioning into drawing the image itself rather than just identifying transformations. Students will be given a sequence of transformations and learn to draw the new image that results after a sequence of transformations is applied to the preimage. Students evaluate tables of values in order to explore a variety of representations in transformations.			
 Component Summary	4.4.1 Warm-Up (~5 min.)	Notice and Wonder about the pattern from a kaleidoscope (<i>SE p. 196</i>)	4.4.3 Guided Instruction (15 min.)	Use coordinates to draw an image as the result of a given sequence of transformations (<i>SE p. 198</i>)
	4.4.2 Collaboration (15 min.)	Draw the transformation of images on a coordinate grid (<i>SE p. 196</i>)	4.4.4 Your Turn! (10 min.)	Use coordinates to draw an image as the result of a given sequence of transformations (<i>SE p. 200</i>)
	4.4.5 Wrap-Up (~5 min.)	Draw an image on the coordinate plane that is the result of a given sequence of transformations (<i>SE p. 201</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Sequence of transformations <u>Additional Terms:</u> - Reflection - Rotation - Translation		(Optional) Patty paper (Optional) Graphing utility (Optional) Colored pencils	
				Additional Preparation N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may swap the x- and y-values with the coordinates when graphing. - Students may reverse the signs when completing rotations. - Students may misunderstand the direction of clockwise and counterclockwise.	Precision matters in this lesson as students begin drawing on a coordinate plane. Rotations and reflections may be hard for students to “see” and even harder to produce on their own. Visit the 4.5.1 Warm-Up for Lesson 5 as a gauge for what students might be able to produce and which elements of precision to highlight and spiral throughout this lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Notice and Wonder The 4.4.1 Warm-Up provides students with a colorful pattern from a kaleidoscope to determine things that they notice and things that they wonder. Consider using responses related to symmetry, reflections, or similar geometric terms as meaningful connections throughout this lesson and unit.	MA.K12.MTR.4.1: Discussion and Reflection 4.4.2 Collaboration provides students with an opportunity to engage in collaborative discussion and reflection with their partner about drawing an image on the coordinate grid given a sequence of transformations.
	For a kinesthetic entry point for all transformations in this lesson, provide copies or patty paper for students to manipulate based on the given transformation. This will allow students to demonstrate their understanding of the transformation prior to drawing it on the grid paper, where mistakes may happen. For example, utilize Desmos or GeoGebra to illustrate each rotation, translation, or reflection in real-time while combining this with student-created foldables or anchor charts for each.	
Lesson Notes:		

4.4.1 Warm-Up: Notice and Wonder

Component Overview: Students will complete a Notice and Wonder activity about the pattern from a kaleidoscope.



4.4.1 Warm-Up: Notice and Wonder

A kaleidoscope is an instrument that contains loose bits of colored material, usually glass, between two flat plates and two plane mirrors. The plates can be moved so the colored material changes position. The mirrors then reflect the material in an endless variety of patterns.



This Photo by Unknown Author is licensed under CC BY

- What do you notice about the pattern from the kaleidoscope in the picture?
Answers will vary. The image has symmetry across any line drawn from the tip of one yellow point through the center to the tip of the yellow point across from it.
- What do you wonder about the pattern?
Answers will vary. Is there symmetry across any lines drawn through the center?



Optional questions to consider if time permits:

- *What geometric shapes do you notice in this image?*
- *How do you think the kaleidoscope creates these shapes without having those shapes on the device?*
- *How do you think it is possible for the colors to appear the way that they do in this image?*
- *Does this make you think any differently about how you view transformations like reflections?*

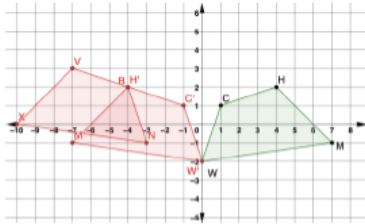
4.4.2 Collaboration: Drawing the Image

Component Overview: Students will work together to draw images on a coordinate grid using their knowledge of transformations. Students should work individually to first draw the transformations before checking their work with their partner.



4.4.2 Collaboration: Drawing the Image

1. Quadrilateral $CHMW$ is shown on the coordinate grid.



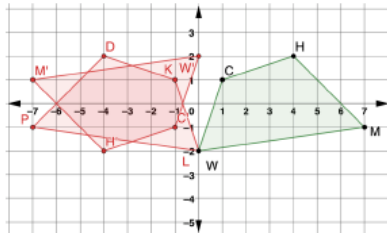
- a. Draw the quadrilateral that is the result of a reflection across the y -axis and then translated left 3 and up 1. Label the new quadrilateral $BVXN$.
Shown on graph.

- b. Check your graph with your partner. Make any necessary adjustments.

A **sequence of transformations**, or a composition of transformations, is when two or more transformations are combined.

- c. Work with your partner to explain why quadrilateral $BVXN$ is a result of a sequence of transformations.
Answers will vary. The quadrilateral is first reflected and then translated, so there is more than one transformation to get the new quadrilateral $BVXN$.

2. Quadrilateral $CHMW$ is shown on the coordinate grid.



This component is designed to lead students through a discovery process. Students may benefit from using colored pencils to show the figure after each transformation.



Consider prompting struggling learners to map one vertex at a time or use patty paper.



Consider the use of colored pencils and highlighters if using a document camera or other display.

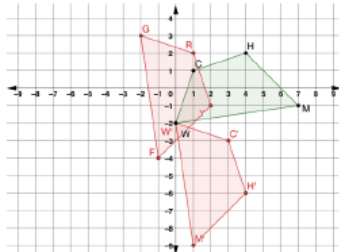
4.4.2 Collaboration: Drawing the Image (continued)

- a. Draw the quadrilateral that is the result of a rotation of 180° counterclockwise about the origin and then reflected across the x -axis. Label the new quadrilateral $KDPL$.

Shown on graph.

- b. Check your graph with your partner. Make any necessary adjustments.

3. Quadrilateral $CHMW$ is shown on the coordinate grid.



- a. Draw the quadrilateral that is the result of a rotation of 90° clockwise about the point W and then translated left 2 and up 5. Label the new quadrilateral $RYFG$.

Shown on graph.

- b. Check your graph with your partner. Make any necessary adjustments.

4. For each sequence of transformations, summarize your mistakes and what you learned from them.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

- a. A reflection across the y -axis and then translated left 3 and up 1
 - b. A rotation of 180° counterclockwise about the origin and then reflected across the x -axis
 - c. A rotation of 90° clockwise about the point W and then translated left 2 and up 5
- Answers will vary.*



Monitor for students or partners with different images and have them compare and contrast their images.



For an extension or to challenge students who finish early, ask students to explain and represent how a figure that is not rotated about the origin could be translated to the origin, rotated, then translated again to “undo” the first translation.

A sample response: Translate the figure by translating the center of rotation, Point W , up two units to be located on the origin. Then follow the algebraic descriptions for rotations listed in Lesson 1. Then translate the rotated figure back down two units.



Allow students to share out their mistakes in a whole-group setting. When student(s) share their mistake(s), poll the class to determine who shares in that mistake(s). This can help students realize that they are not alone in common errors, so they feel encouraged to keep trying. If no student wants to share, list common mistakes you noticed and wrote down during your monitoring of the students' work during the activity. Strategies like this and reflection opportunities like mistakes cultivate a classroom culture of growth mindset and perseverance. Even consider a prize or celebration for students who share the most common mistake.

4.4.3 Guided Instruction: Using Coordinates

Component Overview: Students will receive guided instruction on the use of coordinates to draw an image as the result of a given sequence of transformations. For this activity, consider using the gradual release model of “I Do” and “We Do” throughout.



4.4.3 Guided Instruction: Using Coordinates

The Effect of Transformations on Coordinates

Rotations Clockwise About the Origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
Change the x -coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the y -coordinate to its opposite, then switch the coordinates
Rotations Counterclockwise About the Origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
Change the y -coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the x -coordinate to its opposite, then switch the coordinates

Translations	
Right	Left
Add to the x -coordinate	Subtract from the x -coordinate
Up	Down
Add to the y -coordinate	Subtract from the y -coordinate

Reflections	
x -axis	y -axis
The y -coordinate changes to its opposite.	The x -coordinate changes to its <i>opposite</i> .
$y = x$	$y = -x$
The coordinates switch.	The coordinates switch and become their opposite.



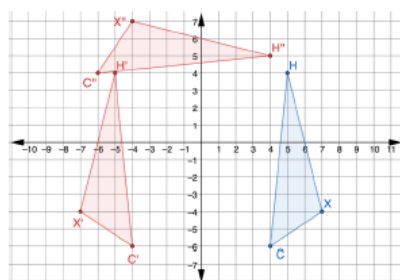
Instruction begins with a written description of the algebraic description of different transformations. To help students connect the algebraic description, have the students write the algebraic description under each written description.



Consider creating this page as a note-catcher sheet or an anchor chart that students can reference or utilize in later lessons throughout the unit, or have students create their own anchor charts personalized to their entry points and learning needs.

4.4.3 Guided Instruction: Using Coordinates (continued)

1. Triangle CHX is shown on the coordinate grid.



Triangle CHX

Vertex	x	y
C	4	-6
H	5	4
X	7	-4

- a. Reflect $\triangle CHX$ across the y -axis then rotate 90° clockwise about the origin. Show both $\triangle C'H'X'$ and $\triangle C''H''X''$.

Shown on graph.

- b. Complete the table of values for each vertex.

Triangle $C'H'X'$

Vertex	x	y
C'	-4	-6
H'	-5	4
X'	-7	-4

Triangle $C''H''X''$

Vertex	x	y
C''	-6	4
H''	4	5
X''	-4	7

- c. Discuss with your partner how the tables show the effect of transformations on coordinates.

Answers will vary. The numbers in the table matched the directions from the guidelines for the given transformation.



- How is the written description of the effect of a reflection across the y -axis seen in the table of values?
- How is the written description of the effect of a rotation of 90° clockwise about the origin seen in the table of values?



For a visual entry point, have students first graph the values from the table. Then encourage students to anticipate where each point will lie (which quadrant) after each translation.

4.4.3 Guided Instruction: Using Coordinates (continued)

One strategy to apply transformations is to use tables of ordered pairs and then to plot the ordered pairs.

2. The table of values shows the coordinates of the vertices of quadrilateral $WBKT$.

Quadrilateral $WBKT$

Vertex	x	y
W	-17	-5
B	-11	-5
K	-4	-11
T	-14	-16

- a. Complete the table of values for the given sequence of transformations of quadrilateral $WBKT$.
Rotate quadrilateral $WBKT$ 180° counterclockwise about the origin, then translate it right 7 down 4.

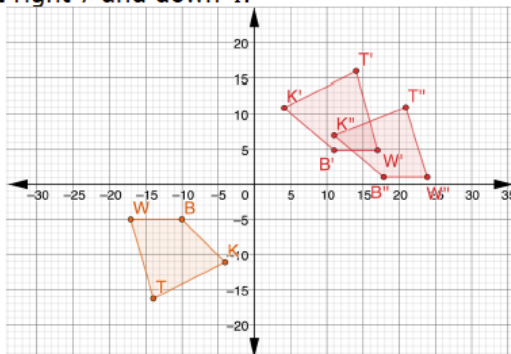
Quadrilateral $W'B'K'T'$

Vertex	x	y
W'	17	5
B'	11	5
K'	4	11
T'	14	16

Quadrilateral $W''B''K''T''$

Vertex	x	y
W''	24	1
B''	18	1
K''	11	7
T''	21	12

- b. Quadrilateral $WBKT$ is shown on the coordinate grid. Show the sequence of transformations on the coordinate grid by rotating quadrilateral $WBKT$ 180° counterclockwise about the origin, then translating it right 7 and down 4.



For students who struggle with tables of values, completing the tables before graphing will be harder. For students who understand transformations and algebraic descriptions, using tables is usually easier to do first before graphing. Question 2 allows students with these strengths to understand that their method is just as effective as graphing. Allow students' choice in which entry point works best for them as they navigate this activity.



Consider using physical manipulatives to assist students with visualizing what this sequence of transformations would look like. For a different visual, consider using a graphing technology such as Desmos or GeoGebra to support their drawings.

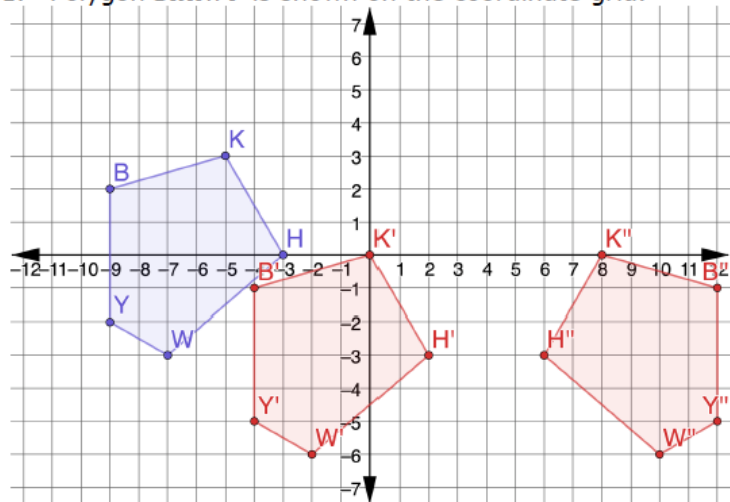
4.4.4 Your Turn!

Component Overview: Students will engage in an independent practice on the use of coordinates to draw an image as the result of a given sequence of transformations. This activity should be done independently to allow students to practice.



4.4.4 Your Turn!

1. Polygon $BKHWY$ is shown on the coordinate grid.



On the coordinate grid, translate polygon $BKHWY$ right 5 and down 3. Then, reflect the result across the line $x = 4$. Label the resulting polygon $B''K''H''W''Y''$.

Shown on graph.



Consider using the time for this activity to work with small groups of students who may need additional support based on their performances during this lesson. Small groups should consist of students who may struggle with transformations at a single step. With these students, review each transformation one at a time, ensuring they understand each part before moving on. During this time, you may determine which students struggle with specific transformations and provide more targeted support.



When reviewing this activity, consider asking students if they used a table of values, mapped the transformations, or use patty paper to help determine the location of the image.

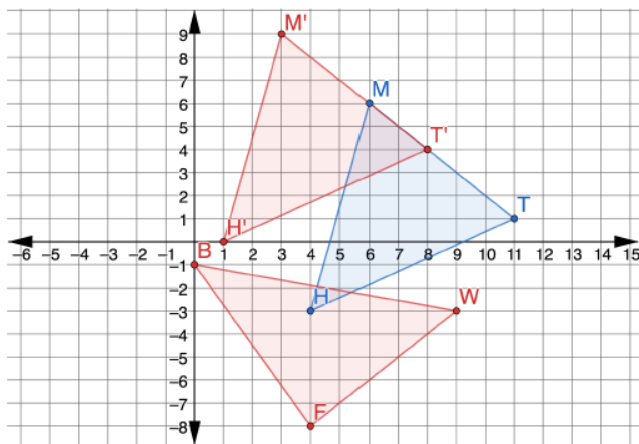
4.4.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door that assesses student understanding of drawing an image on the coordinate plane that is the result of a given sequence of transformations. This activity should be done independently to gather quantitative data on whether or not students understand the content of this lesson.



4.4.5 Wrap-Up: Ticket Out the Door

- Triangle HMT is shown on the coordinate grid.



Draw $\triangle BWFF$, which is the result of translating $\triangle HMT$ left 3 and up 3, then rotating the result 90° clockwise about the origin.

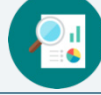
Shown on graph.



For additional insight, consider having students use two different colors to perform each transformation separately. Although this is one sequence of transformations, students drawing each step will provide additional information for future remediation.

4.5 Algebraic Descriptions of a Sequence of Transformations

Lesson Introduction

 Benchmarks	MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.		 Learning Targets	- I can draw an image given a preimage and an algebraic description of a sequence of transformation on a coordinate plane. - I can write a transformation on the coordinate plane using coordinate notation with algebraic descriptors.
 Benchmark Coherence	Building On MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.			N/A
 Essential Question	How can I use the information from an algebraic description of a series of transformations to draw an image?			
 Lesson Narrative	In the lesson, students continue working with drawing an image on the coordinate plane given a preimage and an algebraic description of a sequence of transformations. While students learned this in the prior lesson, students will now be doing the drawing when the sequence of transformations is presented as an algebraic description.			
 Component Summary	4.5.1 Warm-Up (~5 min.)	Write the algebraic description of a given written description (<i>SE p. 202</i>)	4.5.3 Guided Practice (10-15 min.)	Draw the results of a preimage that has gone through a sequence of transformations given as algebraic descriptions (<i>SE p. 204</i>)
	4.5.2 Guided Instruction (15 min.)	Apply a sequence of algebraic descriptions (<i>SE p. 203</i>)	4.5.4 Your Turn! (10 min.)	Draw a quadrilateral that has been transformed using a sequence of transformations presented in algebraic form (<i>SE p. 205</i>)
			4.5.5 Wrap-Up (~5 min.)	Draw a triangle on the coordinate grid that has gone through a sequence of transformations presented in algebraic form (<i>SE p. 206</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Reflection - Rotation - Translation		(Optional) Patty paper	
				Consider writing the coordinates of each transformation on the key in this guide ahead of class for ease in making quick checks to student work during the monitoring step of the lesson.

 Teacher Tips	Common Misconceptions - Students may invert the naming of shapes, use the wrong name, or believe the order does not matter. For example, students may label the 4.5.5 Wrap-Up as triangle <i>SLT</i> instead of triangle <i>TLS</i>. - Students may represent the algebraic description of a reflection of the x-axis as a change in the x-coordinate.	Things to Consider Students may choose a variety of methods for describing, writing, and drawing sequences of transformations. At this point in the unit, they could use a table of values, mapping by vertices, or graphing using descriptions. Provide space and opportunity for students to compare and contrast their methods for collective learning throughout the lesson.
	Literacy and Language Routines Compare and Connect During 4.5.2 Guided Instruction, students apply a sequence of transformations with an algebraic description. Consider a Think Aloud to make explicit connections between the visual representations and algebraic descriptions. Purposefully use language like “Compare this...” or “I am making a connection between.”	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.3.1: Mathematical Fluency Students practice flexibility in their approach to sequences of transformations, using a variety of methods for drawing images using written descriptions, tables of values, or algebraic descriptions. The clarifications in this standard include allowing students to select a procedure, with “efficiency” meaning different things for different students based on their learning entry points.
	 Differentiate Instruction	For a different entry point, consider first breaking the sequence of transformations into three separate problems and having the students draw each separate transformation on separate same-sized grids. Then have the students overlay each transformation to copy all three transformations onto one grid. Making this move will help students who may become overwhelmed by the amount of drawing in one space.
Lesson Notes:		

4.5.1 Warm-Up: Algebraic Descriptions of Transformations

Component Overview: Students will complete a table where they are writing the algebraic description of a given written description. For this activity, allow students to refer to their previously written notes. Encourage students to refine any mistakes they may discover in their notes.



4.5.1 Warm-Up: Algebraic Descriptions of Transformations

- Use your notes to complete the algebraic descriptions in the table.

The Effect of Transformations on Coordinates

a.

Rotations Clockwise About the Origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
Change the x -coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the y -coordinate to its opposite, then switch the coordinates
$(x, y) \rightarrow (y, -x)$	$(x, y) \rightarrow (-x, -y)$	$(x, y) \rightarrow (-y, x)$
Rotations Counterclockwise About the Origin		
Rotations of 90°	Rotations of 180°	Rotations of 270°
Change the y -coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the x -coordinate to its opposite, then switch the coordinates
$(x, y) \rightarrow (-y, x)$	$(x, y) \rightarrow (-x, -y)$	$(x, y) \rightarrow (y, -x)$

b.

Translations	
Right	Left
Add to the x -coordinate	Subtract from the x -coordinate
$(x, y) \rightarrow (x + a, y)$	$(x, y) \rightarrow (x - a, y)$
Up	Down
Add to the y -coordinate	Subtract from the y -coordinate
$(x, y) \rightarrow (x, y + b)$	$(x, y) \rightarrow (x, y - b)$

c.

Reflections	
x -axis	y -axis
The y -coordinate changes to its opposite.	The x -coordinate changes to its opposite.
$(x, y) \rightarrow (x, -y)$	$(x, y) \rightarrow (-x, y)$
$y = x$	$y = -x$
The coordinates switch.	The coordinates switch and become the opposite.
$(x, y) \rightarrow (y, x)$	$(x, y) \rightarrow (-y, -x)$



Notice the similarities and differences between this activity and the synthesis activity *Bring It Together* in Lesson 4. Consider having students use their notes from that activity as a verification step when completing the Warm-Up.

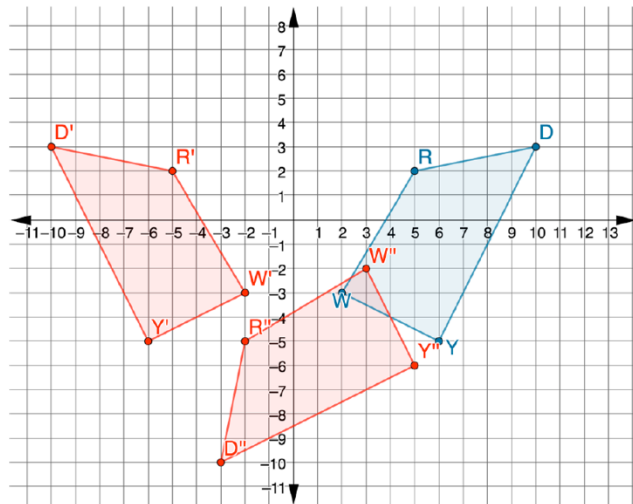
4.5.2 Guided Instruction: Applying Algebraic Descriptions

Component Overview: Students will receive guided instruction on applying a sequence of transformations written as algebraic descriptions. For this activity, leverage the “I Do” and “We Do” components of the gradual release model.



4.5.2 Guided Instruction: Applying Algebraic Descriptions

1. Quadrilateral $RDYW$ is shown on the coordinate grid.



Apply the sequence of transformations to quadrilateral $RDYW$. Draw the result of each transformation on the graph. Assume all rotations are centered at the origin.

- $(x, y) \rightarrow (-x, y)$ Label as $R'D'Y'W'$.
- $(x, y) \rightarrow (-y, x)$ Label as $R''D''Y''W''$.

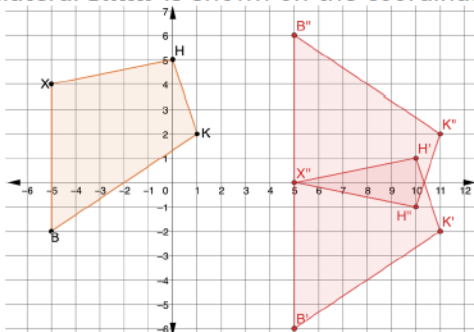
Shown on graph.



If available, consider the use of physical manipulatives or patty paper to help students see each step of the transformation occur as it happens during the sequence.

4.5.2 Guided Instruction: Applying Algebraic Descriptions (continued)

2. Quadrilateral $BXHK$ is shown on the coordinate grid.

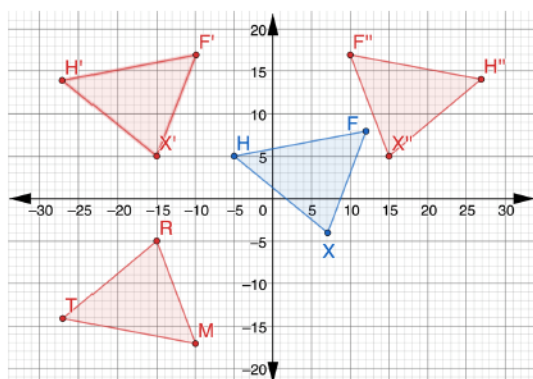


Apply the series of transformations to quadrilateral $BXHK$. Draw the result of each transformation on the graph. Assume all rotations are centered at the origin.

- $(x, y) \rightarrow (x + 10, y - 4)$ Label as $B'X'H'K'$.
- $(x, y) \rightarrow (x, -y)$ Label as $B''X''H''K''$.

Shown on graph.

3. Triangle HFX is shown on the coordinate grid.



Perform the sequence of transformations on $\triangle HFX$. Assume all rotations are centered at the origin. Label the result of all three transformations as $\triangle TMR$.

- $(x, y) \rightarrow (x - 22, y + 9)$
- $(x, y) \rightarrow (-x, y)$
- $(x, y) \rightarrow (-x, -y)$

Shown on graph.



Consider using some or all of the following prompts for inquiry-based learning during this activity:

- What transformation is being applied when you see $(x + 10, y - 4)$? How can you tell? What specifically is asked to be done?
- How would the table of values of the vertices of $BXHK$ be changed when applying $(x + 10, y - 4)$?
- How could you show the transformation of the Vertex K using mapping?
- What transformation is being applied when you see $(x, -y)$? How can you tell? What specifically is asked to be done?
- How would the table of values of the vertices of $B'X'H'K'$ be changed when applying $(x, -y)$?
- How could you show the transformation of the vertex K' using mapping?



When reviewing question 2, consider having the students create a table of values of the vertices of triangle HFX and complete a new table of values for each transformation. Then support students who prefer mapping by showing the mapping of one vertex.



If time permits, allow students to share their answers for question 3 with a partner before showing the class the answer. This will allow students the opportunity to revise their thinking after discussing with their peers before the answers are shared whole group.

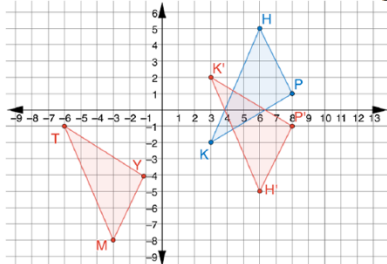
4.5.3 Guided Practice: Applying Algebraic Descriptions

Component Overview: Students will practice with drawing images that are the results of a preimage that has gone through a series of transformations. For this activity, allow students to try on their own and then compare their work with a partner.



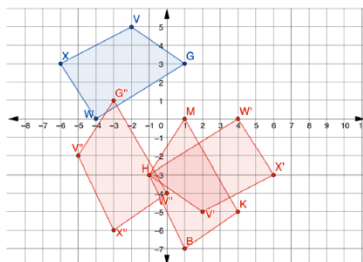
4.5.3 Guided Practice: Applying Algebraic Descriptions

1. Triangle PKH is shown on the coordinate grid.



- a. Perform the sequence of transformations on $\triangle PKH$. Assume all rotations are centered at the origin. Label the result of transformations as $\triangle YTM$.
- $(x, y) \rightarrow (x, -y)$
 - $(x, y) \rightarrow (x - 9, y - 3)$
- Shown on graph.*
- b. Check your answer with your partner.

2. Quadrilateral $XVGW$ is shown on the coordinate grid.



Perform the series of transformations on quadrilateral $XVGW$. Assume all rotations are centered at the origin. Label the result of transformations as quadrilateral $BHMK$.

- $(x, y) \rightarrow (-x, -y)$
- $(x, y) \rightarrow (y, -x)$
- $(x, y) \rightarrow (x + 4, y - 1)$

Shown on graph.



While the primary intent of this activity is an individual activity with a partner review, consider allowing students to work with a partner from the beginning of the Guided Practice. If you do this option for question 1, release students to work on question 2 on their own and then check their answers with their partner.



- Did you and your partner both choose the same method for determining the location of $\triangle YTM$?
- Did you and your partner arrive at the same answer initially? If not, what did you notice that was different?
- Were you able to agree on the final answer?



Consider strategies for breaking down the sequence into three parts representing each step of the sequence or each separate transformation.



Note that three transformations are used in this sequence.



For students who finish early, consider having students create a fourth transformation, displayed algebraically, and add that fourth transformation to their coordinate grid.

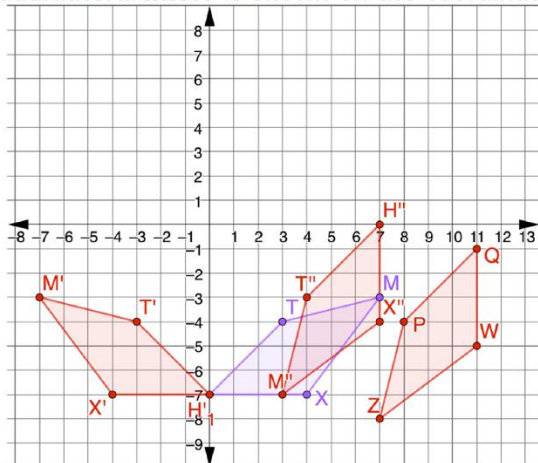
4.5.4 Your Turn!

Component Overview: Students will independently practice drawing a quadrilateral using a sequence of transformations presented in algebraic form.



4.5.4 Your Turn!

1. Quadrilateral $XHTM$ is shown on the coordinate grid.



Perform the sequence of transformations on quadrilateral $XHTM$. Assume all rotations are centered at the origin. Label the result of the transformations as quadrilateral $WQPZ$.

- $(x, y) \rightarrow (-x, y)$
- $(x, y) \rightarrow (-y, x)$
- $(x, y) \rightarrow (x + 4, y - 1)$

Shown on graph.



This activity can be modified for students who may need a challenge or who may need additional scaffolded support.

- *Extension:* Determine if the transformations could be done in the reverse order.
- *Scaffolded:* Reduce the number of transformations down to two.



- *What are the types of transformations being asked for in this question?*
- *How can you tell based on the algebraic description?*

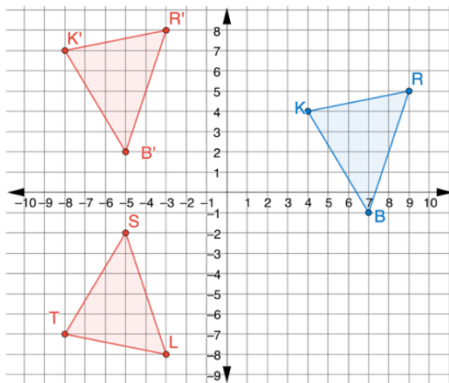
4.5.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door that has students draw a triangle on the coordinate grid that has gone through a sequence of transformations presented in algebraic form. This activity should be done independently to determine student mastery of this lesson's content.



4.5.5 Wrap-Up: Ticket Out the Door

- Triangle KRB is shown on the coordinate grid.



Consider prompting students to provide written descriptions for each transformation, as they may properly identify the type of transformation but still struggle mapping its result.

Perform the sequence of transformations on $\triangle KRB$. Assume all rotations are centered at the origin. Label the result of the transformations as $\triangle TLS$.

- $(x, y) \rightarrow (x - 12, y + 3)$
- $(x, y) \rightarrow (x, -y)$

Shown on graph.

4.6 Isometry

Lesson Introduction

 Benchmarks	MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure. MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.		 Learning Targets	- I can describe a sequence of transformations that map one figure onto another congruent figure. - I can draw an image given a preimage and a sequence of transformations on a coordinate plane.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- How can rigid motions be used to justify that two figures are congruent? - Why do two figures remain congruent once rigid motion is applied?			
 Lesson Narrative	In this lesson, students begin to look at congruency through the lens of rigid motions. Students will begin by exploring transforming parts using multiple coordinate planes and be introduced to the concept of isometry. They will then continue exploring transformations and isometry in order to bridge that knowledge with the formal definition of congruence in terms of rigid motions.			
 Component Summary	4.6.1 Warm-Up (~5 min.)	Which One Doesn't Belong? (<i>SE p. 207</i>)	4.6.3 Exploration (10-15 min.)	Exploring transformations and isometry (<i>SE p. 208</i>)
	4.6.2 Exploration (15 min.)	Exploring transforming parts (<i>SE p. 207</i>)	4.6.4 Bring It Together (5-10 min.)	Synthesis for formal definition of congruence in terms of rigid motions (<i>SE p. 211</i>)
			4.6.5 Wrap-Up (~5 min.)	Growth Mindset survey (<i>SE p. 212</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Isometry - Transformation - Definition of congruence in terms of rigid motions <u>Additional Terms:</u> N/A		- Patty paper (Optional) Graph paper N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not identify parallel arrows during the 4.6.2 Exploration and not preserve elements of distance. - Students may misinterpret the preserving of distance and angle measure as either distance or angle measure.	Exploration and discovery-based learning dominates this lesson. Highlight productive struggle and consider providing strategies for working through that struggle, including sentence stems or partner work as embedded scaffolding throughout each exploration.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Collect and Display During 4.6.2 Exploration and 4.6.3 Exploration, students are engaged with a partner or small group. During this time, consider listening for and scribing the output you are hearing from students using written words, diagrams, and pictures. This can then be organized, revoiced, or explicitly connected to other language in a display for all students to use.	MA.K12.MTR.1.1: Participate in Effortful Learning Learning can feel like a struggle, and each Exploration provides students the opportunity to flex their productive struggle muscles as they engage in discovery-based learning as they explore and apply working definitions of isometry, congruence, and rigid motion. Incorporate clarification language like “perseverance,” “positive mindset,” and “community of learners” into your responses throughout the lesson.
	The use of patty paper during each Exploration provides additional entry points for all learners. Combined with the discovery-based approach of this lesson, students are positioned as experts of their own learning experiences regardless of their initial entry points in the process. Consider having students share their different learning experiences as a strategy for collective learning, building confidence, and highlighting flexibility in learning.	
Lesson Notes:		

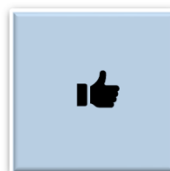
4.6.1 Warm-Up: Which One Doesn't Belong?

Component Overview: Students will review four given images to determine which one does not belong.



4.6.1 Warm-Up: Which One Doesn't Belong?

- Which one doesn't belong? Explain your choice.



The Warm-Up intentionally positions students to attend to precision in the location, orientation, and other features of each image to better prepare for precision in their 4.6.2 and 4.6.3 Explorations.

Answers will vary. The figure on the bottom right does not belong because the hand in this figure looks smaller than the hands in the other figures. The figure in the top left does not belong because it has the only hand that is located in the corner of the figure. The figure in the bottom left does not belong because it is the only figure where the thumb is not pointing up.



After students have chosen one image and explained their responses, challenge them to find at least one other image and provide an explanation for that choice.

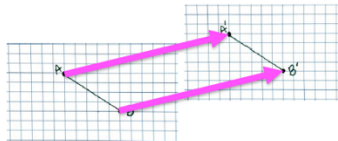
4.6.2 Exploration: Transforming Parts

Component Overview: Students will be doing an exploration with transforming parts. For this activity, students should be working with a partner or small group to go through each of the steps in the process of this activity. At the conclusion of this activity, students will formalize the definition of isometry.



4.6.2 Exploration: Transforming Parts

1. Traci drew $\overline{AB}$ on a piece of graph paper. She then moved the graph paper up and over to the right, as shown by the arrows. The movement created $\overline{A'B'}$.



The graph paper is a real-life example of a plane. When an object is moved, flipped, or rotated, it is the plane that the object is on that moved. This means that every point on the plane is moved, flipped, or rotated.

- a. Draw $\overline{YZ}$ on the same plane as $\overline{AB}$. Then, draw $\overline{Y'Z'}$ on the second plane to show how $\overline{YZ}$ would move in the same way as $\overline{AB}$ when the plane is moved.

Answers will vary in the location of $\overline{YZ}$ which will also yield varying locations of $\overline{Y'Z'}$. Answers should represent a translation. Arrows (vectors/directed line segments) drawn between points Y and Y' should be the same length and go in the same direction as the arrows (vectors/directed line segments) drawn between Z and Z' .

- b. Draw the arrows from the endpoints of $\overline{YZ}$ to the endpoints of $\overline{Y'Z'}$. Compare the arrows from $\overline{YZ}$ to $\overline{Y'Z'}$ to the arrows from $\overline{AB}$ to $\overline{A'B'}$. What do you notice?

Answers will vary. I noticed that the arrows are parallel. Arrows drawn between points Y and Y' are the same length as the arrows drawn between Z and Z' . Arrows point in the same direction.



Provide space and opportunity for productive struggle throughout this Exploration. Students will engage in meaningful learning opportunities by making mistakes or identifying what information they need to know and formulating what questions to ask.



Consider providing students with graph or patty paper in order to replicate what is presented in the narrative of question 1. This may provide a kinesthetic entry point to better see the manipulation of the image.

4.6.2 Exploration: Transforming Parts (continued)

Isometry is a transformation that preserves distance and angle measure.

- c. Use the definition of isometry to justify that Traci's example is an isometry.

Answers will vary. I noticed the distance between points A and B is the same as the distance between points A' and B' . The distance between Y and Z is the same as the distance between Y' and Z' . $\overline{AB}$ is the same length as $\overline{A'B'}$. $\overline{YZ}$ is the same length as $\overline{Y'Z'}$.



"Isometry" will be a new term for students. Question 1c is included intentionally to give students opportunity to extend their working definitions of isometry by making sense of the concept through an example.

4.6.3 Exploration: Transformations and Isometry

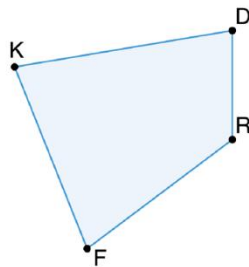
Component Overview: Students will be doing an exploration with transforming parts. For this activity, students should work with a partner or small group to go through each of the steps in the process of this activity. Students will use patty paper to go through each of the steps of this activity.



4.6.3 Exploration: Transformations and Isometry

1. Quadrilateral $KDRF$ is shown.

- a. Use patty paper to copy the length of $\overline{RF}$. Label the endpoints on the patty paper. Flip the patty paper and place it in the space to the left of quadrilateral $KDRF$. Use the patty paper and any other tools to draw the copy of $\overline{RF}$. Name it $\overline{R'F'}$.



Be mindful that this activity requires the use of patty paper to complete. Students may need additional support when it comes to using the patty paper to complete each step. Consider monitoring for opportunities or moments to briefly review student work so that observations or conclusions are more precise and accurate.



A **transformation** is a one-to-one correspondence between two sets of points.

b. Work with your partner to complete the statement:

Since there are a set of points on $\overline{RF}$ and a set of points on $\overline{R'F'}$, it is not enough to just state that the endpoints R and F correspond with endpoints R' and F' because a transformation is a **one-to-one correspondence** between two sets of points. Therefore, to show that $\overline{RF} \cong \overline{R'F'}$, all of the points on $\overline{RF}$ have to have a matching point on $\overline{R'F'}$.



Students may struggle with what to use to complete the blanks. Guide students to refer back to the definition for a clue instead of giving them the answer. Often answers can be found within the content of the lesson. This strategy supports students in becoming more independent learners.

4.6.3 Exploration: Transformations and Isometry (continued)

- c. Use the steps to apply the same transformations to each part of quadrilateral $KDRF$.

- Use patty paper to copy $\angle RFK$. Use the copy of $\angle RFK$ to create $\angle R'F'K'$.
- Use the patty paper and any other tools to draw the copy of $\overline{KF}$. Name it $\overline{K'F'}$.
- Use patty paper to copy $\angle FRD$. Use the copy of $\angle FRD$ to create $\angle F'R'D'$.
- Use the patty paper and any other tools to draw the copy of $\overline{RD}$. Name it $\overline{R'D'}$.
- Use patty paper to copy $\angle FKD$. Use the copy of $\angle FKD$ to create $\angle F'K'D'$.
- Use the patty paper and any other tools to draw the copy of $\overline{KD}$. Name it $\overline{K'D'}$.

- d. Explain why when drawing all of the parts of the quadrilateral $KDRF$ to create quadrilateral $K'D'R'F'$, the movements of the patty paper were the same as when drawing $\overline{R'F'}$.

Answers will vary. The distance between all of the points on the quadrilateral remained the same as it did when making a copy of $\overline{RF}$ to draw $\overline{R'F'}$. To make a copy of quadrilateral $KDRF$, all parts of the new quadrilateral, $K'D'R'F'$, were placed in the same position with respect to each other as the parts of $KDRF$. All points in the new figure were the same distance from each other and placed in the same way, with respect to each other, when making the copy. All the angle measures are the same. Note that quadrilateral $KDRF$ is the preimage, and quadrilateral $K'D'R'F'$ is the image.



Some students will learn more effectively during question 1c by individually engaging with each step, while others will work better articulating and verifying their steps with a partner.

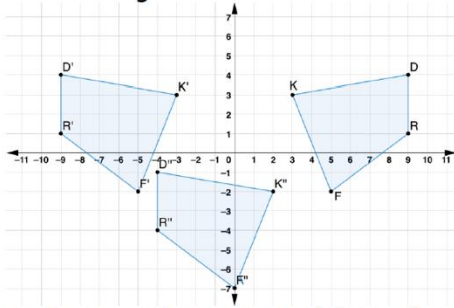


If students are unsure how to answer question 1d, consider using the following questions as prompts:

- *Why do you think the distances between all of the points on the quadrilateral remained the same as they did during making a copy of RF to draw $R'F'$?*
- *Why do all angles stay the same in this case?*
- *Which of the quadrilaterals is the preimage and which is the actual image? Does it matter?*

4.6.3 Exploration: Transformations and Isometry (continued)

2. Quadrilaterals $KDRF$, $K'D'R'F'$, and $K''D''R''F''$ are shown on the coordinate grid.



- Use patty paper to copy quadrilateral $KDRF$.
- Lay the copy of quadrilateral $KDRF$ on top of quadrilateral $KDRF$.
- Describe what could be done so that the copy of quadrilateral $KDRF$ lays on top of quadrilateral $K'D'R'F'$.

Answers will vary. Reflect quadrilateral $KDRF$ across the y -axis.

Reflect quadrilateral $KDRF$ using the line of reflection of $x = 0$.

A horizontal translation of quadrilateral $KDRF$ left 6 units and a reflection with the line of reflection $x = -3$.

- Was the movement of the patty paper an example of isometry?
Yes. When the copy of quadrilateral $KDRF$ was placed on top of quadrilateral $K'D'R'F'$, so that the corresponding points were on top of each other, the distance between the points on quadrilateral $K'D'R'F'$ was the same as the distance between the points on quadrilateral $KDRF$. The angle measures were the same.
- Lay the copy of quadrilateral $KDRF$ on top of quadrilateral $K'D'R'F'$.



When reviewing question 2, consider allowing a student (or student pair) to come to the board or document camera to show the steps from question 2a and 2b.

During the review of question 2c, engage students in a discussion where they are using accurate translation language when completing this process.

Monitor opportunities for more precise academic language in moments when students are describing things like “reflect across the y -axis” as “flip up” or similar informal vocabulary.



Question 2c allows for multiple entry points, as students may see transformed images differently. Allow students to describe the transformations based on what they see and not a predetermined set of moves.

4.6.3 Exploration: Transformations and Isometry (continued)

- f. Describe what should be done next so that the copy of quadrilateral $KDRF$ lays on top of quadrilateral $K''D''R''F''$.

A vertical translation down 5 units and a horizontal translation right 5 units.

- g. Was the movement of the patty paper an example of isometry?

Yes. When the copy of quadrilateral $KDRF$ was placed on top of quadrilateral $K''D''R''F''$, the distance between the points on quadrilateral $K''D''R''F''$ was the same as the distance between the points on quadrilateral $K'D'R'F'$. The angle measures were the same.

- h. Determine if the copy of quadrilateral $KDRF$ could be moved with just one transformation so that it lays on top of quadrilateral $K''D''R''F''$.

No. You would need to perform more than one transformation in order to lay a copy of quadrilateral $KDRF$ on top of quadrilateral $K''D''R''F''$.

3. There is at least one other sequence of transformations that can be used to move the copy of quadrilateral $KDRF$ so that it lays on top of quadrilateral $K''D''R''F''$. Work with your partner to describe one of these sequences of transformations.

Answers will vary. However, the answer to this question depends on the initial placement of the copy of quadrilateral $KDRF$ when performing the first transformation. For instance, to move a copy of quadrilateral $KDRF$ from quadrilateral $KDRF$ to quadrilateral $K''D''R''F''$, performing a transformation of a reflection across $x = 3$ would then require a vertical translation of 5 units down, and horizontal translation of 1 unit left. A different initial placement could be a reflection across $x = 2$, which would then require a vertical translation of 5 units down, and horizontal translation of 1 unit right.



For students working in pairs or groups and stuck on question 2g or 2h, consider the following questions:

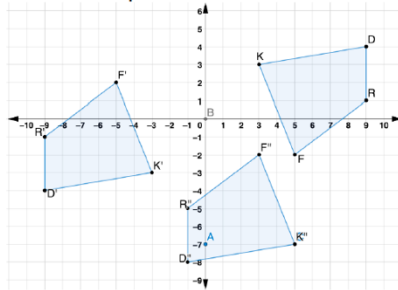
- What conditions did this transformation meet in order to be an example of isometry?
- Why can't a copy of this be moved with just one transformation?



Student responses to question 3 will depend on the initial placement of the copy of the quadrilateral. Monitor for varying responses or approaches to later highlight when reviewing in a whole-group setting.

4.6.3 Exploration: Transformations and Isometry (continued)

4. The coordinate grid shows a different sequence of transformations of quadrilateral $KDRF$.



Describe what movements should be done with your patty paper copy of quadrilateral $KDRF$ so that the copy of quadrilateral $KDRF$ lays on top of quadrilateral $K''D''R''F''$, by first laying on top of quadrilateral $K'D'R'F'$.
*Answers will vary. A rotation counterclockwise 180° with the center of rotation at the point $(0,0)$ then a vertical translation of 4 units down and a horizontal translation of 8 units right.
 A reflection across the y -axis, a reflection across the x -axis, then a vertical translation 4 units down and a horizontal translation 8 units right.*



Monitor students as they close their exploration, using questions like the following to gauge their conceptual understanding.

- How is this sequence of transformations different than the prior?
- Does this sequence of transformations represent isometry? Why or why not?



Consider having students work on question 4 in pairs and then find another pair of students to compare answers with and discuss.

4.6.4 Bring It Together: Definition of Congruence in Terms of Rigid Motions

Component Overview: Students work to formalize the definition of congruence in terms of rigid motions based on the Exploration in the prior activity (4.6.3).



4.6.4 Bring It Together: Definition of Congruence in Terms of Rigid Motions

1. For both sequences of transformations of quadrilateral $KDRF$, the paper copy of quadrilateral $KDRF$ was

- congruent
- not congruent

to

- $K'D'R'F'$ only.
- $K''D''R''F''$ only.
- both $K'D'R'F'$ and $K''D''R''F''$.

The transformations used in both sequences were

- dilations.
- rigid motions.
- dilations and rigid motions.

All of the points of

- $K'D'R'F'$
- $K''D''R''F''$
- $K'D'R'F'$ and $K''D''R''F''$

can be mapped to all

of the points of $KDRF$.



Definition of Congruence in Terms of Rigid Motions

Two figures are congruent if and only if there exists one or more rigid motions, which will map one figure onto the other.

2. Use the definition of congruence in terms of rigid motions to explain how both sequences of transformations for quadrilateral $KDRF$ resulted in quadrilateral $KDRF \cong$ quadrilateral $K''D''R''F''$.

In both sequences of transformations only rigid motions (translations, reflections, or rotations) were used to map $KDRF$ onto $K''D''R''F''$ so the figures are congruent by the definition of congruence in terms of rigid motions



Synthesis is a key component for strategies of Visible Learning (see Unit Guide). Students approach synthesis differently. Consider routines like Three Reads, Think-Pair-Share, or Stronger and Clearer to give students multiple entry points for synthesis in this activity.



Multiselect questions are often seen on high-stakes assessments. Consider a Think Aloud to model strategies or approaches for answering these questions.

For example, begin with the word “congruent” and move to the next word bank to determine a choice that makes sense with congruent, then move forward to each word bank to see if a chain of correct responses can be made to make one correct answer.

4.6.5 Wrap-Up: Growth Mindset

Component Overview: Students will complete a growth mindset survey related to the concepts covered in this lesson.



4.6.5 Wrap-Up: Growth Mindset

The power of “YET.”

“The power of believing that you can improve.” - Carol Dweck

Select all of the concepts that apply to your understanding right now.

Answers will vary.

1. I don’t understand

- isometry
- transformations
- the definition of congruence in terms of rigid motions

YET.

2.

- Isometry
- Mapping
- One-to-one correspondence
- Definition of congruence in terms of rigid motions

doesn’t

make sense to me YET.

3. I can explain

- isometry.
- mapping.
- one-to-one correspondence.
- transformations.
- the definition of congruence in terms of rigid motions.

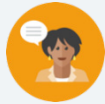


Lesson 7 incorporates a video on growth mindset during the Warm-Up. Consider prompting students to connect their responses here to their working definition of growth mindset and what that can look and sound like for them in this unit.

4.7 Justifying Congruence

Lesson Introduction

 Benchmarks	MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure. MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.		 Learning Targets	- I can describe a sequence of transformations that will map a preimage and an image to show congruence. - I can describe a sequence of transformations that map one figure onto another congruent figure.	
 Benchmark Coherence	Building On MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.			Working Toward N/A	
 Essential Questions	- How can you describe a sequence of transformations that will map a preimage and an image? - How can a sequence of transformations be used to show congruence?				
 Lesson Narrative	Lesson 7 builds on a working definition of congruence to include rigid motion. Students justify that two shapes are congruent (preimage and image) by applying the definition of congruence in terms of rigid motions. Each activity incorporates elements of flexibility, as students can choose which shape is the preimage, providing pathways for a variety of responses.				
 Component Summary	4.7.1 Warm-Up (~5 min.)	Goal-setting activity (<i>SE p. 213</i>)	4.7.3 Guided Practice (10-15 min.)	Practice justifying congruence (<i>SE p. 215</i>)	
	4.7.2 Guided Instruction (15-20 min.)	Justifying congruence through rigid motions (<i>SE p. 213</i>)	4.7.4 Your Turn! (5 min.)	Describe a sequence of transformations using algebraic descriptions (<i>SE p. 216</i>)	
	4.7.5 Wrap-Up (~5 min.)	Ticket Out the Door describing a sequence of transformations to justify congruence (<i>SE p. 217</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> Definition of Congruence in Terms of Rigid Motions <u>Additional Terms:</u> N/A		- Growth Mindset video (Warm-Up) (Optional) Patty paper		4.7.1 Warm-Up contains a video to accompany the question prompt.

 Teacher Tips	Common Misconceptions - Students may recognize the correct transformations but do them in an order that results in the preimage not mapping onto the image. - Students may incorrectly write the transformation algebraically. - Students may not match the corresponding vertices correctly.	Things to Consider The 4.7.1 Warm-Up provides an opportunity to incorporate practices of growth mindset into your lesson. Integrate methods like positive talk, feedback loops, or sentence stems for students to use during each of the activities in the lesson.
	Literacy and Language Routines Stronger and Clearer Each Time During 4.7.2 Guided Instruction, students will justify congruence through rigid motion. Provide students multiple opportunities to articulate their responses, gathering feedback from peers while incorporating that feedback for more refined and precise mathematical answers.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.3.1: Mathematical Fluency Students have the opportunity to identify and share their uniquely different yet correct sequences of transformations in each activity. Because preimage and image are not identified, students' approaches will vary, providing opportunity to position students as experts of their own work, building both confidence and fluency.
	 Differentiate Instruction	
The 4.7.1 Warm-Up provides an opportunity to incorporate practices of growth mindset into your lesson. Integrate methods like positive talk, feedback loops, or sentence stems for students to use during each of the activities in the lesson. - Ask students what James Clear's quote means to them or could mean in a particular situation in their academic life currently. - Create accountability partners for students to identify moments of growth from current results and have students revisit these goals and end results in Unit 5. - Provide sentence stems for students to incorporate positive self-talk into each activity, pausing after each activity to give students 60 seconds to turn and reflect with a partner using positive self-talk.		
Lesson Notes:		

4.7.1 Warm-Up: You Can Do It!

Component Overview: Students will complete an activity focused on setting goals. Consider showing the video prior to having students complete their goals.



4.7.1 Warm-Up: You Can Do It!

"You should be far more concerned with your current trajectory than with your current results." - James Clear

Has anyone ever asked you, "What goals do YOU want to achieve in your lifetime?"



Setting goals is linked to self-confidence, motivation, and autonomy. A 2015 study showed people who wrote down their goals were 33 percent more successful in achieving them than those who just thought about the outcomes. Making goals can be fun.

For example, you can name places you would like to visit, activities you would like to do, or people you would like to meet. By setting the goal, you have already set your trajectory toward achieving it. In the table, write three life goals and 2 goals for the rest of this school year.

Life Goals	Goals for this school year
<i>Answers will vary.</i>	<i>Answers will vary.</i>
"If you are always trying to be normal, you will never know how amazing you can be." Maya Angelou	



Consider building on this activity by having students set goals for this lesson particular lesson, this unit, or this class.

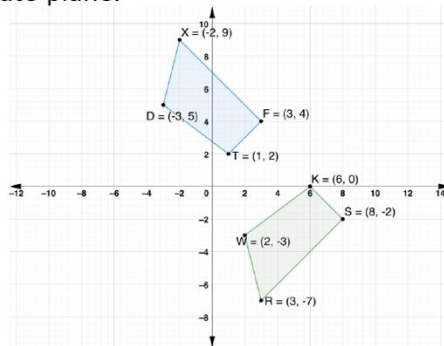
4.7.2 Guided Instruction: Justifying Congruence

Component Overview: Students will describe sequences of transformations that map quadrilaterals onto one another. They will justify congruence of the two quadrilaterals by describing the mapping so that the definition of congruence in terms of rigid motions can be applied. For this activity, employ a gradual release model to guide students through the Guided Instruction.



4.7.2 Guided Instruction: Justifying Congruence

1. Quadrilaterals $XFTD$ and $KSRW$ are shown on the coordinate plane.



- a. Describe a sequence of transformations that will map quadrilateral $KSRW$ onto quadrilateral $XFTD$.
*Answers will vary. A vertical translation down 2 units, a horizontal translation left 5 units, then a reflection across the x -axis.
 A reflection across the line $y = 1$ with a horizontal translation left 5 units.
 A vertical translation up 2 units, horizontal translation left 5 units, then a reflection across the line $y = 2$.*
- b. Complete the table that shows which vertices were mapped to each other when quadrilateral $KSRW$ was mapped onto quadrilateral $XFTD$.

Quadrilateral $KSRW$	Quadrilateral $XFTD$
K	T
S	F
R	X
W	D



Consider an alternate entry point of having students create a table of values for each quadrilateral. Ask the students to informally describe the transformations that they think occurred. Then focus on one vertex and apply the transformations using the ordered pairs or map them on the grid.



How do you determine which vertex from one quadrilateral corresponds to which vertex from the other? Why would the congruence statement be wrong if incorrect vertices were mapped?

4.7.2 Guided Instruction: Justifying Congruence (continued)



Definition of Congruence in Terms of Rigid Motions

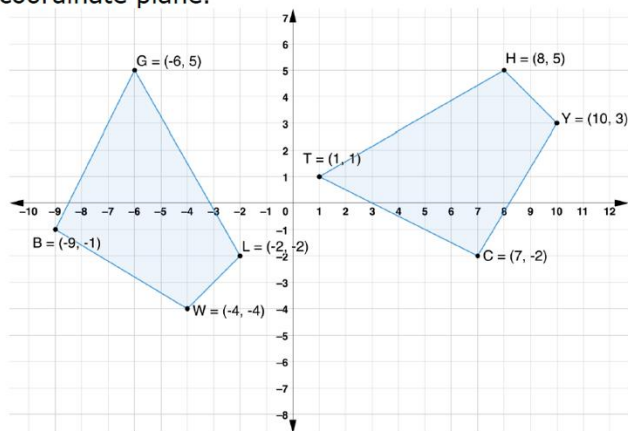
Two figures are congruent if and only if there exists one or more rigid motions, which will map one figure onto the other.

- c. Complete the statements.

Quadrilateral $KSRW$ is congruent to quadrilateral $XFTD$ because the sequence of transformations that can be used to map the quadrilaterals were **rigid motions**.

Because vertex K maps to vertex T , vertex S maps to vertex F , vertex R maps to vertex X and vertex W maps to vertex D . The definition of congruence in terms of rigid motions justifies that $KSRW \cong TFXD$.

2. Quadrilaterals $BGLW$ and $HYCT$ are shown on the coordinate plane.



Stronger and Clearer

The formal definition of congruence in terms of rigid motions was introduced in the prior lesson. Consider giving students an opportunity to make sense of information using the example and prompt. Have students incorporate an “if and only if” meaning within their responses.



For question 1c, refer to the definition to point out the conditions of the definition of congruence in terms of rigid motions were met. Avoid stating that congruence exists because the figures are the same shape and the same size.



Consider an inquiry-based approach to Guided Instruction. What transformations do you think occurred with quadrilaterals $BGLW$ and $HYCT$? Which quadrilateral did you use as the preimage? How could choosing the other quadrilateral as the preimage change the transformations?

4.7.2 Guided Instruction: Justifying Congruence (continued)

- a. Use algebraic descriptions to describe a sequence of transformations that will map quadrilateral $BGLW$ onto quadrilateral $HYCT$.

Answers will vary.

$$(x, y) \rightarrow (-y, x)$$

$$(x, y) \rightarrow (x + 6, y + 7)$$

OR

$$(x, y) \rightarrow (-y, -x)$$

$$(x, y) \rightarrow (x, -y)$$

$$(x, y) \rightarrow (x + 6, y + 7)$$

- b. Complete the table that shows which vertices were mapped to each other when quadrilateral $BGLW$ was mapped onto quadrilateral $HYCT$.

Quadrilateral $BGLW$	Quadrilateral $HYCT$
B	C
G	T
L	H
W	Y



When using rigid motions to map one figure onto the other, always be sure the corresponding vertices match before stating congruence.

- c. Complete the congruence statement.

quadrilateral $BGLW \cong$ *quadrilateral* $CTHY$



MA.K12.MTR.3.1: Mathematical Fluency

Notice the flexible pathways for responses since the question does not specify preimage and image. Provide opportunity for students to build flexibility in their approaches by monitoring for and sharing student responses that are different.



- Why would being sure that the corresponding vertices match before stating congruence be an important step in this process?
- Does the order the vertices are listed matter when writing a congruence statement?

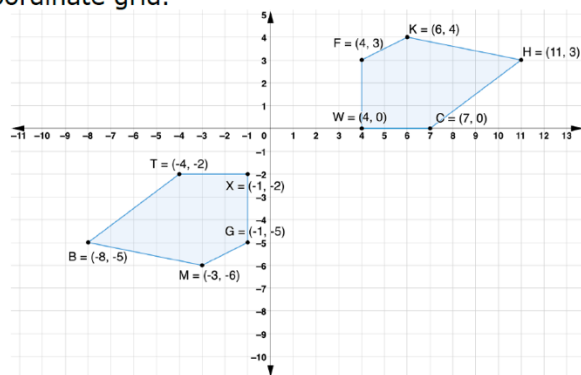
4.7.3 Guided Practice: Justifying Congruence

Component Overview: Students will engage in sequences of transformations to a preimage onto an image. They will practice using this to justify congruence of the two figures. For this activity, release students to work with a partner to practice these questions.



4.7.3 Guided Practice: Justifying Congruence

1. Polygons $TXGMB$ and $WFKHC$ are shown on the coordinate grid.



- a. Use algebraic descriptions to describe a sequence of transformations that will map polygon $TXGMB$ onto polygon $WFKHC$.

Answers will vary.

$$(x, y) \rightarrow (-x, -y)$$

$$(x, y) \rightarrow (x + 3, y - 2)$$

OR

$$(x, y) \rightarrow (-x, y)$$

$$(x, y) \rightarrow (x, -y)$$

$$(x, y) \rightarrow (x + 3, y - 2)$$

- b. Complete the congruence statement.

polygon $TXGMB \cong$ *polygon CWFKH*



Consider doing a *Think Aloud* to launch this activity. This can help students better understand what to look for when preparing to complete the activity. Allow students to share the transformations they see, making connections between visual and verbal descriptions and their algebraic representations.



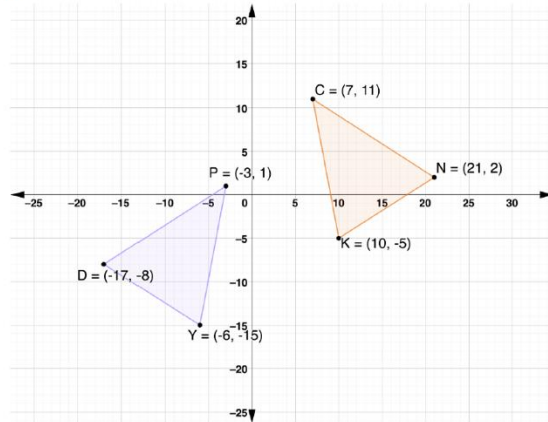
Ensure students understand that there can be multiple sequences of transformations that show congruence.



Monitor for and have students share work that highlights different methods (tables of values, mapping, written descriptions) that can be used to write the algebraic descriptions.

4.7.3 Guided Practice: Justifying Congruence (continued)

2. Triangles CKN and PYD are shown on the coordinate grid.



- a. The sequence of transformations that will map $\triangle CKN$ onto $\triangle PYD$ is

- ☐ translating $\triangle CKN$ left 14,
- ☐ reflecting $\triangle CKN$ across the y -axis,
- ☐ rotating $\triangle CKN$ 270° clockwise about the origin,

then

- ☐ translating $\triangle C'K'N'$ right 4, down 10.
- ☐ translating $\triangle C'K'N'$ right 10, down 4.
- ☐ reflecting $\triangle C'K'N'$ across the x -axis.

- b. Vertex C maps to vertex P , vertex K maps to vertex Y , and vertex N maps to vertex D therefore $\triangle CKN$ is congruent to $\triangle PYD$ by the definition of congruence in terms of rigid motions.



In question 2a, were you able to quickly eliminate any of the options in the first box? In the second box? Explain your rationale.



For a challenge, have students attempt to find a way to make one of the incorrect answers correct using information not provided as one of the initial answer choices.

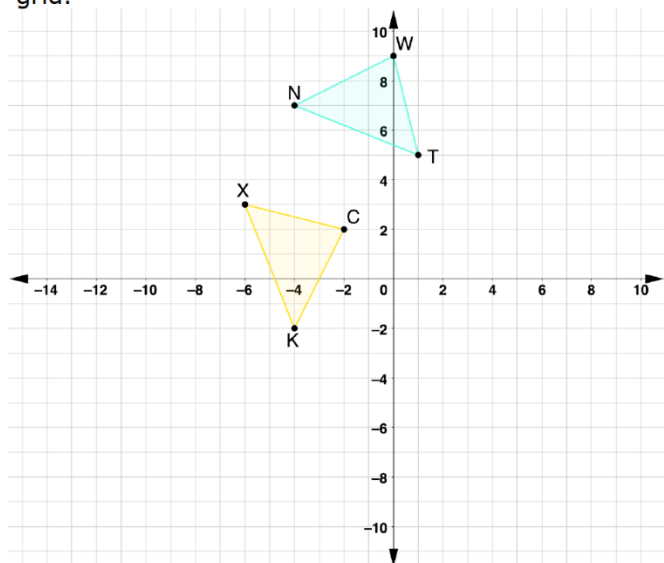
4.7.4 Your Turn!

Component Overview: Students will work independently to write algebraic descriptions to describe a sequence of three transformations that will map a triangle onto another.



4.7.4 Your Turn!

1. Triangles XCK and TWN are shown on the coordinate grid.



Use algebraic descriptions to describe a sequence of three transformations that will map $\triangle XCK$ onto $\triangle TWN$.

Transformations
$(x, y) \rightarrow (-x, y)$
$(x, y) \rightarrow (y, -x)$
$(x, y) \rightarrow (x - 2, y + 11)$



Consider including a “my favorite no” to highlight creative thinking that can highlight common misconceptions and provide opportunity for clarification.



Consider allowing students to answer with a written/verbal description of the transformations to check their understanding of the rigid motions. Then use the tables of the algebraic descriptions from previous lessons to provide support in making connections between these descriptions to the algebraic descriptions.



To add a layer of rigor to this activity, consider having students switch the preimage and image and write a new sequence of transformations.

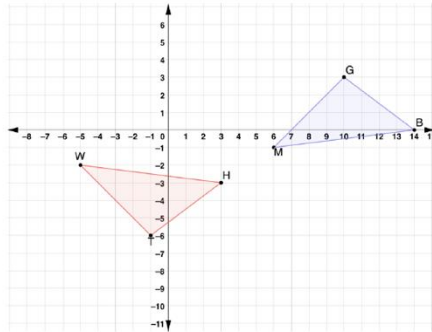
4.7.5 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door on describing a sequence of transformations to justify congruence. This activity should be done independently to collect student-level performance data.



4.7.5 Wrap-Up: Ticket Out the Door

- Triangles TWH and GBM are shown on the coordinate grid.



- Use written descriptions or algebraic descriptions to describe a sequence of two transformations that will map the two triangles onto each other.

*To map $\triangle GBM$ onto $\triangle TWH$, then perform a reflection across the x -axis, $(x, y) \rightarrow (x, -y)$ and a horizontal translation left 11 units and vertical translation down 3 units, $(x, y) \rightarrow (x - 11, y - 3)$.
To map $\triangle TWH$ onto $\triangle GBM$, then perform a reflection across the x -axis, $(x, y) \rightarrow (x, -y)$ and a horizontal translation right 11 units and vertical translation down 3 units $(x, y) \rightarrow (x + 11, y - 3)$.*

- Complete the table that shows which vertices were mapped to each other.

$\triangle TWH$	$\triangle GBM$
T	G
W	M
H	B

- Complete the congruence statement.
 $\triangle TWH \cong \triangle GMB$



Notice this question intentionally omits the identification of preimage and image, as either triangle could be the preimage for the congruence statement to still be true.



Consider prompting students who finish early to determine if multiple sequences could map the same images or how changing which triangle is the preimage would change the sequence of transformations.

4.8 Putting It All Together

Lesson Introduction

 Benchmarks	MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure. MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.		 Learning Targets	- I can determine a sequence of transformations that map one figure onto another congruent figure. - I can match the written and algebraic descriptions of a sequence of transformations given the preimage and image of a figure.
 Benchmark Coherence	Building On MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.		Working Toward N/A	
 Essential Question	How can I apply what I have learned in this unit to determine a sequence of transformations that map one figure onto another congruent figure?			
 Lesson Narrative	This lesson serves as a culminating experience for this unit. Students will apply all that they have learned throughout the unit to complete a card sort where they will match a coordinate grid that shows the preimage and image of a triangle and match it with its appropriate written description and algebraic description.			
 Component Summary	4.8.1 Warm-Up (~5 min.)	Notice and Wonder of a diagram activity (<i>SE p. 218</i>)	4.8.2 Activity (30-35 min.)	Card Sort activity with matching a coordinate grid with the correct written description and algebraic description (<i>SE p. 218</i>)
4.8.3 Wrap-Up (~5 min.)	Self-reflection activity on current learning for the learning targets for this lesson (<i>SE p. 219</i>)			
 Resources	Glossary		Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		Go through the Card Sort activity on your own prior to having students do it. This will help you look for potential errors or mismatches that may occur. By doing this, you can quickly observe where a group may be going wrong while facilitating.	
		Materials		
		Card sort activity cards		

 Teacher Tips	Common Misconceptions • Students may misinterpret the preimage vs the image in the coordinate grid cards. • Students may mistake their signs when looking at the algebraic description cards. • Students may misread the written description cards in this activity.	Things to Consider • This lesson is primarily the 4.8.2 Activity Card Sort. It will require physical materials (the cards for the activity) to be prepared for students prior to the lesson. Consider having this done before the lesson to allow more time for the actual activity itself.
	Literacy and Language Routines Notice and Wonder The 4.8.1 Warm-Up leverages the Notice and Wonder protocol. Encourage low-floor answers like the difference in the type of grid while also prompting students to extend their answers for high-ceiling statements that include learning from the unit like rigid motions, isometry, or congruence.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations In the 4.8.2 Activity, students will be completing a Card Sort where they match a transformation provided in the coordinate grid (preimage and image), written description, and algebraic description.
	 Differentiate Instruction Consider the placement of students in their small groups based on successes observed with each of the individual transformations in the previous lessons of this unit. This will allow students an entry point of using their individual strengths during the Card Sort activity. Also consider eliminating some of the cards for a shorter activity. Suggested cards for eliminating are card 5, card 6, card 9, and card 10. Consider using the Card Sort with a group of students who need further support in small group instruction.	
Lesson Notes:		

4.8.1 Warm-Up: Notice and Wonder

Component Overview: Students will engage in a Notice and Wonder activity, examining a quadrilateral that has been reflected and translated.



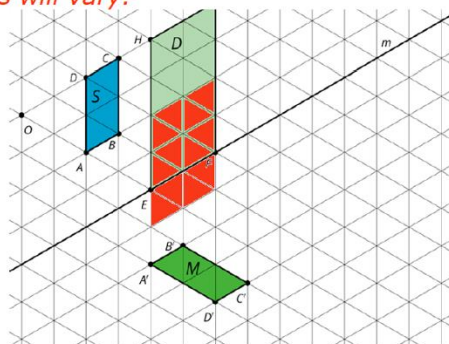
4.8.1 Warm-Up: Notice and Wonder

1. What are three things you notice about the diagram?

Answers will vary. I noticed that the three quadrilaterals are different sizes, they are different colors and two of them are the same size. I noticed that image D has 16 triangles while M and S have 4, D and M are shades of green while S is blue, M uses prime notation and S and D do not.

2. Reflect quadrilateral $EFGH$ across line m , then translate it up 3 units. Name the resulting quadrilateral $E'F'G'H'$.

Answers will vary.



3. Check your location of quadrilateral $E'F'G'H'$ with your partner. If your locations differ, explain why both could be correct. If your locations are the same, find another possible interpretation of up 3 units and draw that quadrilateral.

Answers will vary. My partner and I had the same answer but "up three" could have gone to the right instead. My partner reflected the same but went to the right instead of going vertically.



The Notice and Wonder intentionally provides opportunity for students to explore isometric graphing and its usefulness. Prompt students to notice visual differences while extending some student responses to include the usefulness of isometric paper when showing congruence. Encourage productive struggle and partner talk.

4.8.2 Activity: Card Sort

Component Overview: Students will work collaboratively with a partner or small groups to match graphed transformation, written description, and algebraic description cards that represent the same sequence of transformations. For this activity, consider reviewing the instructions and logistics of the activity prior to launching. Leave time in this activity for a review of the correct answer and sharing of the processes involved.



4.8.2 Activity: Card Sort

Match each coordinate grid with the preimage of $\triangle TLW$ and image $\triangle HPY$ where $\triangle TLW \cong \triangle HPY$, with the written and algebraic descriptions that map $\triangle TLW$ onto $\triangle HPY$. Use the letters in the blue, green, and orange boxes to record your matches in the table.

Coordinate Grid	Written Description	Algebraic Description
B	W	AA
D	N	BB
F	J	EE
G	K	GG, HH, or YY
H	S	CC
M	V	ZZ
T	L	GG, HH, or YY
W	C	FF
X	Q	GG, HH, or YY
Y	P	DD



Consider modeling one example for instructions. For example, show that a “match” is a graphed transformation, a written description card, and an algebraic description card where all three represent the same sequence of transformations. Also, students should know that no cards will be used more than once and none will be left over.



Consider eliminating some of the cards for a shorter activity. Suggested cards for eliminating are card 5, card 6, card 9, and card 10. Consider giving patty paper or other manipulatives for students to use to engage kinesthetically with the physical transformations of the images.



Engage students in a discussion after the card sort before moving to the next component. As part of the discussion, ask students how they determined the solutions.



For a challenge, remove some of the written description and algebraic description cards and replace them with blank cards where the students must write the description that is missing for that set.

4.8.3 Wrap-Up: Reflection

Component Overview: Students will complete a self-reflection where they indicate where their current learning is related to the learning targets of this lesson. This activity should be done independently to allow for individual responses.



4.8.3 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is in relation to each target.

Answer will vary.

1. I can match the written descriptions of a sequence of transformations to the coordinate grid with the pre-image and image.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.



Refer to the Unit Guide for a digital, interactive extension on writing algebraic descriptions for transformations.

2. I can match the algebraic descriptions of a sequence of transformations to the coordinate grid with the pre-image and image.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

3. I can determine a sequence of transformations that map one figure onto another congruent figure.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.